

SUPPLEMENT: DOPE: DOWNSTREAM OBJECTIVE-PRESERVING ENCODING FOR SYNTHETIC TABULAR DATA

Anonymous

Affiliation withheld for review.

ABSTRACT

Lineage-level retention and byte charges for the displayed profile, the 8,192-step sufficiency budget, and the 142 BeyondArena family names. These tables are the rows behind the medians in the main paper. A blank retention is an unavailable fit or an auditor that is not informative on all three sample seeds. It is not a zero. A lineage with no charged bytes and no retention is omitted. BeyondArena is not pooled with the PMLB panel. The release-score rows are arithmetic on invented components. They are not measurements on these lineages.

A DISPLAYED PROFILE, ALL 100 LINEAGES

Table 1: Displayed profile, all 100 lineages. The cap column is charged bytes at or under 10,240. It is not a release certificate.

id	name	rows	feat.	bytes	cap	CB n	CB $4n$	lin $4n$	MLP $4n$
02a45777900441ce	215_2dplanes	800	10	2,583	pass	0.9930	1.0001	0.9990	1.0477
047e62b52a3f4ede	695_chatfield_4	156	12	3,311	pass	1.0041	1.0155	1.0048	—
04c80944e971cb3e	first_principles_supernovae_zr	157	1	762	pass	0.8753	0.8846	0.9959	1.2627
0561d673cb73c27e	666_rmftsa_ladata	339	10	2,921	pass	1.0095	1.1026	1.0338	1.1394
060b9fff8092db17	589_fri_c2_1000_25	667	25	5,717	pass	0.8893	0.9534	0.9911	1.5794
06b0312a585b436e	1196_BNG_pharynx	800	10	2,631	pass	0.8601	0.8964	1.0458	1.7049
0779d3fd5b3133a2	nikuradse_2	241	1	758	pass	0.9096	0.9261	2.1723	2.2082
07f38946316c039b	593_fri_c1_1000_10	667	10	2,876	pass	0.5650	0.6068	0.9617	1.6005
0a8428ef25a69175	537_houses	800	8	2,609	pass	0.6257	0.7036	1.0150	1.0256
0d482c1dd7cc6592	210_cloud	72	5	1,844	pass	1.0374	0.9772	1.0944	1.0239
0db59fd2579600d0	665_sleuth_case2002	98	6	1,962	pass	1.0371	1.1324	1.1345	—
0f86a714ec5f3ac6	1191_BNG_pbc	800	18	4,183	pass	0.8097	0.8761	0.9173	—
167578f5deba4977	547_no2	333	7	2,443	pass	0.8073	0.8513	1.0097	0.9898
19f2c3c2d67defd7	feynman.II.11_27	800	4	1,494	pass	0.9704	0.9890	0.9960	1.0048
19f4780b53b3fa41	678_visualizing_environmental	74	3	1,515	pass	—	—	—	0.4431
1a4456d1cc71a26a	feynman.test.1	800	7	2,191	pass	1.0608	1.1671	0.9816	—
1a803b27c2400dc4	feynman.III.10_19	800	4	1,520	pass	0.9882	0.9952	0.9942	0.9965
1cb97b1d7314d43d	229_pwLinear	133	10	2,587	pass	0.8417	0.9878	0.9104	—
205f4b4982b9ea49	608_fri_c3_1000_10	667	10	2,845	pass	0.5263	0.5671	1.0451	0.8182
26eff843be35d831	581_fri_c3_500_25	333	25	5,647	pass	0.7602	0.9020	1.1297	1.5792
282f8ddb43b5e677	595_fri_c0_1000_10	667	10	2,904	pass	0.8974	0.9234	0.9962	1.0250
297b7688a045c583	603_fri_c0_250_50	167	50	10,171	pass	0.8848	1.1562	1.0700	2.6883
2c8d0985f3557d2b	feynman.II.11_28	800	2	1,026	pass	0.9831	0.9899	1.0031	0.9976
30ca2887550f0d8d	201_pol	800	26	8,145	pass	0.5742	0.5976	0.7015	0.6923
3408a6a513c627a3	strogatz_barmag1	267	2	1,011	pass	0.9750	0.9775	0.9969	1.0058
35a9aa9b88f0e4d1	344_mv	800	10	2,737	pass	0.9985	1.0016	0.9662	1.0347
3d3321e27d14f9aa	599_fri_c2_1000_5	667	5	1,707	pass	0.8359	0.8611	0.9668	1.6797
44eaabe4cb923766	613_fri_c3_250_5	167	5	1,712	pass	0.8671	0.9276	0.8599	—
46b25db9396c8209	227_cpu_small	800	12	3,327	pass	0.8569	0.9072	0.9018	0.9364
46c5a59e5ae42037	1193_BNG_lowbwt	800	9	2,436	pass	0.9635	0.9703	1.0030	0.9871
471f11cc2d38853a	feynman.test.11	800	4	1,481	pass	0.9645	0.9760	0.9975	0.9826
47c581af5de1136b	582_fri_c1_500_25	333	25	5,670	pass	0.7510	0.9088	0.7173	—
48b7a3bf81e5f3e5	605_fri_c2_250_25	167	25	5,692	pass	0.4046	0.6465	1.9138	—
4a354d98c965b01f	597_fri_c2_500_5	333	5	1,713	pass	0.7550	0.8055	0.9814	1.8381
4aa84fff8ad0cd26	feynman.III.13_18	800	4	1,501	pass	0.8748	0.9137	0.8850	0.9348
51730b348b1bbd14	557_analcatdata_apnea1	316	3	1,237	pass	-0.0032	0.0086	—	—
54b221668673164c	feynman.III.14_14	800	5	1,726	pass	1.0346	1.1164	1.0011	1.0611
552665868f8f72ed	strogatz_vdp2	267	2	989	pass	0.9945	0.9963	0.9960	0.9969
585fef2a1cb6b33f	strogatz_lv2	267	2	1,018	pass	0.8286	0.9859	0.7502	—
642738a361f1c63a	feynman.I.12_1	800	2	1,032	pass	0.9892	0.9910	0.9892	0.9933

id	name	rows	feat.	bytes	cap	CB n	CB $4n$	lin $4n$	MLP $4n$
64c64f0f9983c48f	strogatz_bacres2	267	2	998	pass	0.9793	0.9832	0.9783	0.9862
654566a1f91408aa	4544_GeographicalOriginalofMusic	706	117	22,233	fail	—	—	—	—
693890aae9dc860b	first_principles_supernovae_zg	162	1	761	pass	0.8492	0.8908	1.0066	1.1997
6cacb11611345a24	feynman_I.11.19	800	6	1,947	pass	0.9524	0.9812	0.9780	0.9800
7291b347fe0238fb	feynman_II.11.20	800	5	1,757	pass	0.9928	1.0432	1.0016	1.0674
7423783f7e38bb9b	strogatz_shearflow1	267	2	1,021	pass	0.4900	0.5614	-0.3978	—
7a3d123dd555110d	598_fri_c0.1000.25	667	25	5,668	pass	0.8613	0.9166	1.0046	1.0034
7c8943b160464add	601_fri_c1.250.5	167	5	1,719	pass	0.7859	0.8552	1.0773	—
7e51711baa1b1fb1	feynman_I.12.11	800	5	1,718	pass	0.9267	0.9320	1.0015	0.9860
80589368fa78f21a	218_house.8L	800	8	2,392	pass	0.5835	0.6708	0.9616	1.1040
8164950b155f7f56	feynman_I.12.2	800	4	1,511	pass	0.9702	0.9986	0.9485	0.9753
85a31ae409944a1b	feynman_II.10.9	800	3	1,269	pass	0.9330	0.9508	0.9589	0.9515
8d859cfa7b51a351	feynman_III.15.12	800	3	1,248	pass	0.5172	0.5542	1.0228	1.0749
923483635406b213	1203_BNG_pwLinear	800	10	2,581	pass	1.2079	1.1750	0.9941	1.5230
9312d813229df035	690_visualizing_galaxy	216	4	1,458	pass	0.9725	0.9755	0.9853	1.0870
94a00ea67f49bf84	596_fri_c2.250.5	167	5	1,698	pass	0.7546	0.8072	—	—
94bd4520b8ceb3f0	feynman_test.12	800	5	1,747	pass	0.9688	0.9804	0.9913	1.0067
956d34f13e7af830	586_fri_c3.1000.25	667	25	5,701	pass	0.9325	0.9849	1.2339	2.1457
9a10c15885022034	.deprecated.195_auto.pric	106	15	5,025	pass	0.8717	0.9485	0.9598	1.3107
a04f964bc53281e0	strogatz_lv1	267	2	985	pass	-3.8784	-3.4212	—	—
a23f4bd626b76590	feynman_test.13	800	5	1,740	pass	0.9178	0.9605	0.9900	1.0991
a32a69e7c701eb14	588_fri_c4.1000.100	667	100	19,019	fail	—	—	—	—
a5ba682572f53add	602_fri_c3.250.10	167	10	2,900	pass	0.5355	0.8109	0.7988	—
a863cd42be4d05d4	663_rabe.266	80	2	1,014	pass	0.9781	0.9933	0.9967	1.0521
a97065df07a47eb0	592_fri_c4.1000.25	667	25	5,655	pass	0.8791	0.9340	0.9578	1.2714
ab279eded3b3c52b	574_house.16H	800	16	4,035	pass	0.1181	0.1796	0.1795	0.3705
aebf5d861a8365b4	522_pm10	333	7	2,409	pass	0.5082	0.6149	0.4125	2.3076
b6adf79118a58cea	562_cpu_small	800	12	3,353	pass	0.6968	0.7747	0.9834	0.9307
b78d323566930bd7	583_fri_c1.1000.50	667	50	10,098	pass	0.8607	0.9444	0.9960	1.6211
b7f9774e2142a758	strogatz_shearflow2	267	2	1,019	pass	0.4707	0.5439	0.7450	1.8955
bc85856c012bcc1b	607_fri_c4.1000.50	667	50	10,209	pass	0.8856	0.9640	3.9394	—
c107ee39eb333a9d	584_fri_c4.500.25	333	25	5,638	pass	0.8062	0.8619	0.9417	—
c30ff9405c916718	579_fri_c0.250.5	167	5	1,692	pass	0.9475	1.0923	0.9961	0.9783
c4f8e81b70d33a13	590_fri_c0.1000.50	667	50	10,180	pass	0.8839	0.9172	1.0359	1.1762
c649a4cee87c4321	529_pollen	800	4	1,506	pass	1.0356	1.1012	0.9717	1.2459
cb5bf66ef2ee09ed	nikuradse.1	241	2	992	pass	0.9346	0.9373	0.9997	1.1909
cba8393879afd1dc	230_machine.cpu	140	6	2,338	pass	1.0026	1.0891	1.1718	—
ce1a79e2d658eb8a	strogatz_glider2	267	2	1,003	pass	0.6382	0.6625	1.0720	1.3756
ce837ccd24a3a18c	strogatz_bacres1	267	2	1,004	pass	0.7352	0.8309	0.9014	0.9826
cee6f811d0bf199d	feynman_II.11.3	800	5	1,728	pass	0.9385	0.9666	0.8950	1.0149
d255d3a2e1c8dbf4	feynman_III.12.43	800	2	1,028	pass	0.9920	0.9935	0.9966	0.9940
d52bfd2b6ab1c850	feynman_I.10.7	800	3	1,250	pass	0.9920	0.9932	0.9940	0.9959
d9a3408a840c9f48	560_bodyfat	168	14	4,364	pass	1.0247	1.0436	0.9930	1.3651
da290fae0d6c072f	feynman_test.10	800	3	1,267	pass	0.9955	0.9990	0.9990	1.0015
da78606950f100cd	1199_BNG_echoMonths	800	9	2,544	pass	0.9962	1.0086	0.9655	1.0694
dd16ac872ffb8673	strogatz_glider1	267	2	991	pass	0.1145	0.1723	0.9123	—
de286ac837699fd	564_fried	800	10	2,861	pass	0.9477	0.9829	0.9543	1.4707
e08f0b9931f93686	225_puma8NH	800	8	2,439	pass	1.0639	1.1043	0.9951	2.8498
e18350fb51a9d1b0	strogatz_barmag2	267	2	1,027	pass	0.8897	0.9399	0.2855	—
e5a3a7b3668a1cc2	556_analcatdata.apnea2	316	3	1,238	pass	0.0339	0.0341	—	—
e68fd96cc63271b	573_cpu_act	800	21	4,954	pass	0.8080	0.9045	0.6593	1.1465
ed2efb82fce4b098	strogatz_vdp1	267	2	984	pass	0.5228	0.5812	1.2867	1.2401
ede9d41a298fd60	612_fri_c1.1000.5	667	5	1,705	pass	0.8224	0.8596	0.9670	1.2891
efd53e9c0f36d608	.deprecated.207_autoPrice	106	15	5,017	pass	0.9020	0.9357	1.0899	1.0398
f0a3d6881458f990	604_fri_c4.500.10	333	10	2,913	pass	0.6868	0.8360	0.8552	—
f24e9383331c43e2	503_wind	800	14	3,846	pass	0.9690	0.9962	1.0017	1.0367
f652ea96eff62654	strogatz_predprey1	267	2	989	pass	0.9964	1.1345	1.0218	1.4876
f8cf362e371e8960	strogatz_predprey2	267	2	1,030	pass	0.9876	0.9847	0.9718	1.0815
fc3812567b5fdd64	197_cpu_act	800	21	5,011	pass	0.7680	0.8922	0.6669	1.1253
fed67ac3fcd3129e	1201_BNG_breastTumor	800	9	2,427	pass	-0.6141	-0.3437	—	—

B SUFFICIENCY BUDGET

Table 2: Sufficiency budget. 69 lineages have a charged byte count or a retention. 31 lineages have neither and are omitted. The cap column is charged bytes at or under 10,240. It is not a release certificate.

id	name	rows	feat.	bytes	cap	CB n	CB $4n$	lin $4n$	MLP $4n$
047e62b52a3f4ede	695_chatfield.4	156	12	3,353	pass	1.0106	1.0287	1.0110	—
04c80944e971cb3e	first_principles_supernovae_zr	157	1	750	pass	0.9372	0.9454	1.0346	1.3271
060b9fff8092db17	589_fri_c2.1000.25	667	25	5,734	pass	0.9237	0.9886	1.0258	1.5430
06b0312a585b436e	1196_BNG_pharynx	800	10	2,681	pass	0.8188	0.8768	1.0632	0.7968
07f38946316c039b	593_fri_c1.1000.10	667	10	2,869	pass	0.9780	1.0222	0.9929	1.8097
0a8428ef25a69175	537_houses	800	8	2,636	pass	0.7243	0.7841	1.0073	1.0334
0d482c1dd7cc6592	210_cloud	72	5	1,855	pass	0.9561	0.8815	1.0056	0.9224

id	name	rows	feat.	bytes	cap	CB n	CB $4n$	lin $4n$	MLP $4n$
0f86a714ec5f3ac6	1191_BNG_pbc	800	18	4,218	pass	0.8069	0.8576	0.8783	—
167578f5deba4977	547_no2	333	7	2,474	pass	0.8821	0.9276	0.9850	1.0276
1a4456d1cc71a26a	feynman_test.1	800	7	2,185	pass	1.1162	1.3026	0.6721	—
1a803b27c2400dc4	feynman_III.10_19	800	4	1,522	pass	0.9698	0.9777	0.9732	0.9807
1cb97b1d7314d43d	229_pwLinear	133	10	2,598	pass	0.8275	1.0471	0.9727	—
26eff843be35d831	581_fri.c3_500_25	333	25	5,655	pass	0.8660	0.9960	1.0900	1.6067
282f8ddb43b5e677	595_fri.c0_1000_10	667	10	2,913	pass	0.9645	1.0058	1.0029	1.1269
297b7688a045c583	603_fri.c0_250_50	167	50	10,183	pass	0.8257	1.1292	1.0240	2.8999
30ca2887550f0d8d	201_pol	800	26	8,215	pass	0.6214	0.6436	0.7350	0.6603
3d3321e27d14f9aa	599_fri.c2_1000_5	667	5	1,713	pass	0.9820	1.0044	1.0441	1.6148
44eaabe4cb923766	613_fri.c3_250_5	167	5	1,708	pass	0.8605	0.9611	0.9427	—
46b25db9396c8209	227_cpu_small	800	12	3,338	pass	0.7033	0.8730	0.9142	0.9908
46c5a59e5ae42037	1193_BNG_lowbwt	800	9	2,468	pass	0.8393	0.8390	0.8798	0.8103
471f11cc2d38853a	feynman_test.11	800	4	1,472	pass	0.9340	0.9481	0.9730	0.9578
47c581af5de1136b	582_fri.c1_500_25	333	25	5,676	pass	0.8232	0.9820	0.8086	—
48b7a3bf81e5f3e5	605_fri.c2_250_25	167	25	5,696	pass	0.2608	0.5247	1.4268	—
4a354d98c965b01f	597_fri.c2_500_5	333	5	1,715	pass	0.9578	1.0001	1.0224	1.3200
51730b348b1bbd14	557_analcatdata_apnea1	316	3	1,246	pass	0.6556	0.6654	—	—
54b221668673164c	feynman_III.14_14	800	5	1,724	pass	1.0175	1.1226	1.0021	1.0727
585fef2a1cb6b33f	strogatz_lv2	267	2	1,021	pass	0.9239	1.0451	0.8041	—
642738a361f1c63a	feynman_L.12_1	800	2	1,034	pass	0.9936	0.9961	0.9993	0.9989
64c64f0f9983c48f	strogatz_bacres2	267	2	996	pass	0.9743	0.9826	0.9816	0.9923
693890aae9dc860b	first_principles_supernovae_zg	162	1	766	pass	0.9415	0.9642	1.0277	1.1876
6cacb11611345a24	feynman_L.11_19	800	6	1,952	pass	0.9911	1.0176	1.0047	1.0089
7423783f7e38bb9b	strogatz_shearflow1	267	2	1,023	pass	0.4860	0.5973	-4.3352	—
7a3d123dd555110d	598_fri.c0_1000_25	667	25	5,672	pass	0.8570	0.9206	1.0010	1.0168
7e51711baa1b1fb1	feynman_L.12_11	800	5	1,720	pass	0.9780	0.9927	0.9910	1.0559
80589368fa78f21a	218_house_8L	800	8	2,410	pass	0.6661	0.7528	0.6830	1.0728
85a31ae409944a1b	feynman_II.10_9	800	3	1,264	pass	0.9803	0.9961	0.9881	1.0061
8d859cfa7b51a351	feynman_III.15_12	800	3	1,280	pass	0.7428	0.8572	0.8771	1.2306
9312d813229df035	690_visualizing_galaxy	216	4	1,470	pass	0.9770	0.9875	0.9843	1.0661
94a00ea67f49bf84	596_fri.c2_250_5	167	5	1,718	pass	0.8262	0.8934	—	—
94bd4520b8ceb3f0	feynman_test.12	800	5	1,745	pass	0.9677	0.9845	0.9937	1.0157
9a10c15885022034	.deprecated_195_auto_price	106	15	5,038	pass	0.8965	0.9544	0.9901	1.3165
a04f964bc53281e0	strogatz_lv1	267	2	986	pass	-3.0531	-2.4502	—	—
a32a69e7c701eb14	588_fri.c4_1000_100	667	100	19,041	fail	—	—	—	—
a5ba682572f53add	602_fri.c3_250_10	167	10	2,902	pass	0.7774	0.9773	0.9471	—
a863cd42be4d05d4	663_rabe_266	80	2	1,014	pass	0.9849	0.9996	1.0018	1.0573
ab279eded3b3c52b	574_house_16H	800	16	4,076	pass	0.0805	0.1211	-0.7472	-0.3199
aebf5d861a8365b4	522_pm10	333	7	2,450	pass	0.6372	0.7959	0.6203	3.4124
b78d323566930bd7	583_fri.c1_1000_50	667	50	10,098	pass	0.8817	0.9737	0.8733	0.9465
b7f9774e2142a758	strogatz_shearflow2	267	2	1,022	pass	0.7388	0.8339	-0.4873	1.5681
c107ee39eb333a9d	584_fri.c4_500_25	333	25	5,641	pass	0.8307	0.9024	0.9623	—
c30ff9405c916718	579_fri.c0_250_5	167	5	1,700	pass	0.9081	1.0750	0.9949	0.9830
c649a4cee87c4321	529_pollen	800	4	1,506	pass	1.0619	1.1198	0.9983	1.2801
cb5bf66ef2ee09ed	nikuradse.1	241	2	990	pass	0.9662	0.9677	0.9970	1.1953
ce1a79e2d658eb8a	strogatz_glider2	267	2	1,004	pass	0.5396	0.6099	1.1474	1.2017
ce837ccd24a3a18c	strogatz_bacres1	267	2	1,004	pass	0.7807	0.8568	0.8858	0.9570
d255d3a2e1c8dbf4	feynman_III.12_43	800	2	1,023	pass	0.9947	0.9964	0.9984	0.9967
d52bfd2b6ab1c850	feynman_L.10_7	800	3	1,248	pass	0.9978	0.9988	0.9994	1.0005
da290fae0d6c072f	feynman_test.10	800	3	1,257	pass	0.9681	0.9718	0.9737	0.9744
da78606950f100cd	1199_BNG_echoMonths	800	9	2,578	pass	0.9473	0.9673	0.9447	1.0028
de2868ac837699fd	564_fried	800	10	2,864	pass	0.9826	1.0201	0.9548	1.5094
e08f0b9931f93686	225_puma8NH	800	8	2,440	pass	1.0168	1.0664	0.9433	2.7529
e5a3a7b3668a1cc2	556_analcatdata_apnea2	316	3	1,241	pass	0.6282	0.6707	—	—
e68fdd96cc63271b	573_cpu_act	800	21	4,958	pass	0.8507	0.9689	0.5661	1.1177
ed2efb82fce4b098	strogatz_vdp1	267	2	991	pass	0.5397	0.6472	-0.1705	1.1154
efd53e9c0f36d608	.deprecated_207_autoPrice	106	15	5,024	pass	0.9035	0.9215	1.0781	1.0481
f0a3d6881458f990	604_fri.c4_500_10	333	10	2,918	pass	0.8895	1.0064	1.0421	—
f24e9383331c43e2	503_wind	800	14	3,877	pass	0.9992	0.9953	0.9918	1.0034
f8cf362e371e8960	strogatz_predprey2	267	2	1,030	pass	0.9865	0.9827	0.9658	1.0750
fc3812567b5fdd64	197_cpu_act	800	21	5,006	pass	0.8164	0.9529	0.6632	1.1726

C BEYONDARENA FAMILIES IN THE DATASET BUCKET

Table 3: All 142 BeyondArena families.

name	task	rows	cols	license	split
5g_energy_consumption	regression	92,629	22	MIT	grouped
acquire_valued_shoppers_challenge	binary	160,057	112	not cleared	temporal
airfoil_self_noise	regression	1,503	6	CC-BY-4.0	iid
allstate_claims_severity	regression	188,317	131	not cleared	iid
amazon_employee_access	binary	32,769	10	public-domain	iid
amex_non_iid	binary	1,249,605	191	not cleared	grouped

Table 3, continued

name	task	rows	cols	license	split
anes_voting_2026	binary	48,587	319	not cleared	temporal
aps_failure	binary	76,000	171	CC-BY-4.0	iid
asp_potassco_classification	multiclass	1,212	138	not cleared	grouped
audiology_diagnosis	multiclass	199	69	CC-BY-4.0	iid
bad_customer_detection	binary	1,723	14	public-domain	iid
bank_customer_churn	binary	10,000	11	public-domain	iid
bank_marketing	binary	45,211	14	CC-BY-4.0	iid
biogeographical_ancestry_prediction	multiclass	635	105	not cleared	iid
biomechanical_orthopaedic_prediction	multiclass	310	7	CC-BY-4.0	iid
bioresponse	binary	3,751	1777	public-domain	iid
blood_tests_drink_prediction	regression	345	6	CC-BY-4.0	iid
blood_transfusion	binary	748	5	CC-BY-4.0	iid
body_density_prediction	regression	252	14	not cleared	iid
california_house_prices_2020	regression	41,528	42	not cleared	temporal
cardiotocography	multiclass	2,126	24	CC-BY-4.0	grouped
churn	binary	5,000	20	public-domain	iid
cirrhosis_patient_survival_prediction	regression	161	18	not cleared	iid
climate_model_weather_forecasting	regression	1,250,000	101	not cleared	temporal
clock_protein_toxicity	binary	171	1118	CC-BY-4.0	iid
coffee_rating_prediction	regression	2,369	13	not cleared	temporal
coil_2000	binary	9,822	86	CC-BY-4.0	iid
concrete_compressive_strength	regression	1,030	9	CC-BY-4.0	iid
consumer_complaints	multiclass	1,226,140	13	us-gov	temporal
cooking_time	regression	1,250,000	197	not cleared	temporal
covertypes	multiclass	512,625	15	CC-BY-4.0	grouped
credit_approval	binary	690	16	CC-BY-4.0	iid
credit_card_clients_default	binary	30,000	24	CC-BY-4.0	iid
credit_g	binary	1,000	21	CC-BY-4.0	iid
customer_satisfaction_in_airline	binary	129,880	22	public-domain	iid
delivery_eta	regression	1,250,000	226	not cleared	temporal
dementia_prediction	multiclass	370	10	not cleared	grouped
diabetes_130_us	binary	71,518	45	CC-BY-4.0	iid
diamonds	regression	53,940	10	MIT	iid
drug_induced_autoimmunity_prediction	binary	597	178	CC-BY-4.0	iid
early_learning_predictors	regression	18,874	745	CC-BY-4.0	grouped
early_stage_diabetes_risk_prediction	binary	251	17	CC-BY-4.0	iid
ecoli_proteins	multiclass	327	7	CC-BY-4.0	iid
ecommerce_shipping	binary	10,999	11	public-domain	iid
electric_motor_temperature_prediction	regression	1,296,316	111	not cleared	grouped
emscad	binary	17,460	18	CC0-1.0	iid
erythematous_squamous_disease	multiclass	366	35	CC-BY-4.0	iid
fiat_500	regression	1,538	8	CC0-1.0	iid
fitness_club	binary	1,500	7	public-domain	iid
food_delivery_time	regression	45,451	10	not cleared	iid
forensic_glass_identification	multiclass	214	10	CC-BY-4.0	iid
forest_fires	regression	517	13	CC-BY-4.0	iid
gallstone_disease	binary	319	39	CC-BY-4.0	iid
garments_worker_productivity	regression	1,197	16	CC-BY-4.0	temporal
ghanas_indigenous_intel	multiclass	10,928	11	not cleared	temporal
give_me_some_credit	binary	150,000	11	not cleared	iid
hazelnut_spread_contaminant_detection	binary	2,400	31	not cleared	iid
healthcare_insurance_expenses	regression	1,338	7	not cleared	iid
heart_disease_cleveland	binary	303	14	CC-BY-4.0	iid
heart_disease_hungary	binary	294	14	CC-BY-4.0	iid
heart_disease_va_long_beach	binary	200	14	CC-BY-4.0	iid
heart_failure_followup_survival	binary	299	13	CC-BY-4.0	iid
heloc	binary	10,459	24	not cleared	iid
hepatitis_c_prediction	multiclass	608	13	CC-BY-4.0	iid
hepatitis_survival_prediction	binary	155	20	CC-BY-4.0	iid
hiva_agnostic	binary	3,845	1519	public-domain	iid
home_credit_default_risk	binary	307,507	505	not cleared	iid
home_credit_default_stability	binary	1,224,927	712	not cleared	temporal
homeq_default_prediction	binary	5,708	13	not cleared	iid
homesite_quote_conversion	binary	260,753	296	not cleared	iid
horse_colic_survival	multiclass	344	21	CC-BY-4.0	iid
hotel_booking_demand	binary	81,418	32	CC-BY-4.0	temporal
houses	regression	19,675	9	not cleared	iid
hr_analytics	binary	19,158	13	CC0-1.0	iid
ieee_fraud_detection	binary	590,540	436	not cleared	temporal
immoscout_german_house_prices	regression	10,317	24	not cleared	iid
in_vehicle_coupon_recommendation	binary	12,684	25	CC-BY-4.0	iid
indian_liver_patient_dataset	binary	583	11	CC-BY-4.0	iid
iranian_churn	binary	2,850	14	CC-BY-4.0	iid
jm1	binary	10,885	22	public-domain	iid
kdd_cup_09_appetency	binary	50,000	213	public-domain	iid
kick	binary	72,983	33	public-domain	temporal
kickstarter	binary	187,118	16	not cleared	temporal
labour_inspection_compliance	binary	63,634	377	CC0-1.0	iid

Table 3, continued

name	task	rows	cols	license	split
lending_club	binary	1,064,751	97	CC0-1.0	temporal
ljubljana_breast_cancer	binary	286	10	CC-BY-4.0	iid
ljubljana_primary_tumor	multiclass	302	18	CC-BY-4.0	iid
lung_cancer	multiclass	197	12601	not cleared	iid
lung_cancer_epithelial_genexp	binary	187	22216	public-domain	iid
maps_router_eta	regression	1,250,000	989	not cleared	temporal
marketing_campaign	binary	2,240	26	CC-BY-4.0	iid
maternal_health_risk	multiclass	1,014	7	CC-BY-4.0	iid
mercari_price_suggestion	regression	1,250,000	7	not cleared	iid
mercedes_benz_greener_manufacturing	regression	4,204	372	not cleared	temporal
miami_housing	regression	13,776	16	not cleared	iid
mic	multiclass	1,699	112	CC-BY-4.0	iid
mice_protein_trisomy_discriminant	multiclass	1,080	78	CC-BY-4.0	grouped
micro_mass	multiclass	571	1084	CC-BY-4.0	grouped
musk	binary	6,598	168	CC-BY-4.0	grouped
mutual_funds_india	regression	793	13	CC0-1.0	iid
naticusdroid_android_permissions_dataset	binary	7,491	86	CC-BY-4.0	iid
obesity_estimation	regression	498	15	CC-BY-4.0	iid
online_shoppers_purchasing_intention_dataset	binary	12,330	18	CC-BY-4.0	iid
otto_group_product_classification_challenge	multiclass	61,878	94	not cleared	iid
pancreatic_cancer_mouse_detection	binary	181	6773	not cleared	grouped
parkinsons_biomedical_voice_measurements	binary	195	25	CC-BY-4.0	grouped
physiochemical_protein	regression	45,730	10	CC-BY-4.0	iid
polish_companies_bankruptcy	binary	5,790	65	CC-BY-4.0	iid
porto_seguro	binary	595,206	38	not cleared	iid
predict_students_dropout_and_academic_success	multiclass	4,424	37	CC-BY-4.0	iid
prostate_cancer_detection	binary	322	15155	not cleared	iid
pva_revenue_prediction_kddcup98	binary	144,095	478	not cleared	iid
qsar_aquatic_toxicity	regression	546	9	CC-BY-4.0	iid
qsar_biodeg	binary	1,054	42	CC-BY-4.0	iid
qsar_fish_toxicity	regression	908	7	CC-BY-4.0	iid
qsar_tid_11	regression	5,741	1025	public-domain	iid
regensburg_pediatric_appendicitis	binary	763	52	not cleared	iid
rossmann_store_sales	regression	844,392	16	not cleared	temporal
santander_customer_satisfaction	binary	71,080	308	not cleared	iid
santander_customer_transaction_prediction	binary	200,000	601	not cleared	iid
santander_transaction_value	regression	4,447	541	not cleared	iid
sat11_hand_algo_runtime	regression	2,960	171	not cleared	grouped
sberbank_housing_market_forecasting	regression	27,195	387	not cleared	temporal
sdss_17	multiclass	78,053	12	public-domain	iid
seismic_bumps	binary	2,584	16	CC-BY-4.0	iid
sepsis_prediction	binary	1,228,686	44	not cleared	grouped
sepsis_survival_minimal_clinical_records	binary	110,204	4	CC-BY-4.0	iid
sf_permit_time	regression	116,954	38	not cleared	temporal
south_africa_coronary_heart_disease	binary	462	10	not cleared	iid
splice	multiclass	3,190	61	CC-BY-4.0	iid
student_portuguese_performance	regression	649	31	CC-BY-4.0	iid
superconductivity	regression	21,263	82	CC-BY-4.0	iid
taiwanese_bankruptcy_prediction	binary	6,819	93	CC-BY-4.0	iid
telemonitoring_parkinsons_biomedical_voice_measurements	regression	502	21	CC-BY-4.0	grouped
thyroid_discordant	binary	3,711	27	CC-BY-4.0	iid
tour_travels_churn	binary	954	7	CC0-1.0	iid
video_game_fps_prediction	regression	12,288	40	CC-BY-4.0	grouped
video_transcoding_time_prediction	regression	68,784	20	CC-BY-4.0	grouped
website_phishing	multiclass	1,353	10	CC-BY-4.0	iid
wids_diabetes_mellitus	binary	127,358	182	not cleared	iid
wine_quality	regression	6,497	13	CC-BY-4.0	iid
wine_world_cost	regression	1,279	15	CC0-1.0	iid

D REPLAY CURVES

The main paper reports the median hidden-basis loss. The curves below are the logged replay of both step budgets, before the readout refit. The 8,192-step panel is not the displayed model. The solid line is the training median and the dashed line is the validation median. Each band is a 95% lineage-bootstrap interval. All 100 lineages finished. Median training loss starts at 0.321 and ends at 0.00274 and 0.000763. Median validation loss starts at 0.271 and ends at 0.00443 and 0.00181. Where both budgets have a complete three-seed group, the linear medians are 0.990 and 0.987 (64 lineages; difference -0.000283). The MLP medians are 1.065 and 1.057 (52 lineages; difference 0.00451).

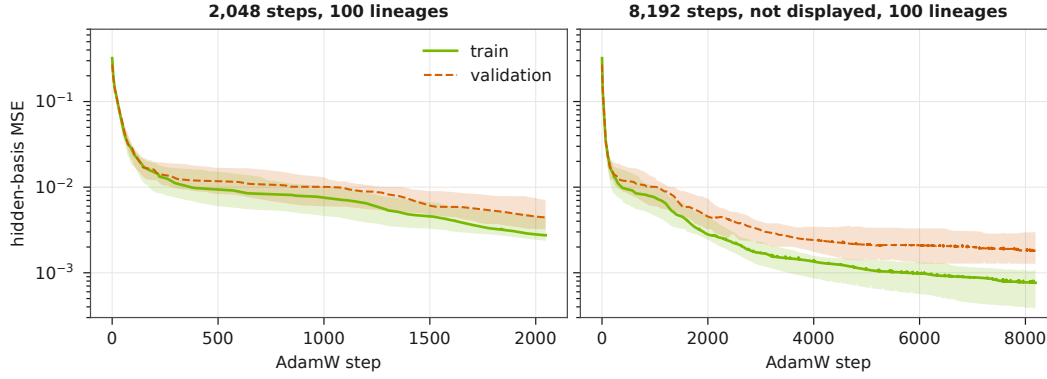


Figure 1: Logged replay of both step budgets before the readout refit. The solid line is the training median and the dashed line is the validation median. Each band is a 95% lineage-bootstrap interval. The longer budget is not the displayed model.

Table 4: Invented applications of the bounded MFS-v3 formula. These are arithmetic illustrations.

Invented input	Scalar
All seven components equal to one	100
Compactness component set to zero	50.12
Byte gate failed	null

E AUTHENTICATED FIRST-BATCH DIAGNOSTICS

The canonical-job batch is incomplete at population and sample-group level. The following values are individual physical evaluations. Logical aliases share one evaluation and its cost. DOPE jobs in this batch use the frozen research profiles named in the source receipt; they do not replace the displayed profile. No population interval or ranking is computed. Pearson entries are the pinned source mean; alpha and beta use its pinned unembedded variant, and MIA is an empirical distance attack. Per-column marginals, all detection folds, DCR/NNDR, applicability reasons, costs and full job digests are in the first-batch source listed in `docs/whitepaper/generated/retained-evidence.json`. The operation-clock sum excludes training and sampling. Official tests remain sealed and all gated scores remain null.

Job prefix	Method	Pearson diff.	Alpha	Beta	Distance MIA AUC
0002af084a92	independent_marginals	0.805	0.602	0	0.474
000377f02a18	DOPE	0.071	0.836	0.558	0.504
000f66aa8642	DOPE	0.266	0.876	0.187	0.595
001897658278	Chow-Liu	0.194	0.804	0.450	0.632
0021656cfc57	Forest-Flow	0.046	0.780	0.555	0.614
00514b7a27e2	DOPE	0.143	0.875	0.239	0.517
0056ef8c748b	independent_marginals	0.149	0.696	0.361	0.637
006a72169115	GaussianCopula	0.064	0.878	0.383	0.452
006cfa72e1a	DOPE	0.069	0.838	0.725	0.710
006dfe86527f	independent_marginals	0.072	0.805	0.491	0.478
007f4ac594ee	TVAE	0.134	0.944	0.523	0.615
0090be69a85e	GaussianCopula	0.098	0.867	0.480	0.393

F RELEASE SCORE THAT WAS NOT EMITTED

MFS-v3 is a gated geometric mean. Every observed scalar remains null. Copying a retention number into that scalar would publish a value the contract refuses to emit. Table 4 applies the bounded MFS-v3 implementation to invented all-ones components and to the same profile with compactness set to zero. The all-ones row is exactly 100. At the byte ceiling, compactness is zero and the formula uses its epsilon floor. A failed byte gate returns null. These rows illustrate arithmetic only.

G REPRODUCIBILITY

Regenerate every `generated/` file and the figure PDFs with `just paper` from the commit that contains these files. The same recipe writes the journal PDF, the anonymous journal PDF, the supplement PDF, and the anonymous supplement PDF. That recipe reads the committed validation ledgers, original fit metadata, and compressed numeric replay logs. The replay cost, loss curves, and BeyondArena extracts have file and field mappings in `generated/original-field-map.json`. Those original metadata snapshots were captured after the historical runs; they leave historical custody and the scored artifacts in place. Pinned LICENSE extracts are reread when their local store is available. Official test files stay sealed. The catalog hash is `ab9fda8d2dea46067b70a42812e9d3c1d9dc6ba025780100df34377e81aa1120`. The grouped training split uses seed 1729. The primary displayed DOPE and legacy comparison fits use seed 11 and sample seeds 101, 211, and 307. Later retained panels document additional ARF and Forest-Flow fit seeds and the measured TabSyn and TabDDPM subsets separately. `python3 research/benchmark/verify_paper_numbers.py` exits nonzero when a displayed macro disagrees with those ledgers. Full neural baseline coverage, the pre-specified privacy-attack panel, displayed-DOPE fit-seed variance, and the full ablation grid stay named in the prose and stay out of the tables until those runs exist. The journal fidelity table is a separate empirical ledger on the grouped validation split. It stands apart from the pre-specified privacy-attack panel. Empirical resemblance stays outside differential privacy and outside HIPAA de-identification.

PMLB is MIT. Every dataset entry in the data lock records SPDX MIT and the note that the source license is MIT (PMLB). CTGAN and TVAE share one pinned LICENSE file: Business Source License 1.1, licensor DataCebo, Inc., licensed work CTGAN. The Gaussian copula uses the pinned Copulas LICENSE: Business Source License 1.1, licensor DataCebo, Inc., licensed work Copulas. TabSyn is Apache License 2.0. TabDDPM is MIT License, Copyright (c) 2022 Akim Kotelnikov. Forest-Diffusion is MIT License, Copyright (c) 2023 Samsung Electronics Co., Ltd. All Rights Reserved. ARF is MIT License, Copyright (c) 2023 Kristin Blesch, Marvin Wright. Chow-Liu and the independent marginals are implementations written for this comparison, under this repository's license: MIT License, Copyright (c) 2026 Dope contributors. The SDV repository LICENSE is absent from the pinned snapshot, so this paragraph leaves that license unrecorded.

Peak RAM in the compute table is resident bytes, shown in mebibytes to one decimal, and peak VRAM is `peak_gpu_used_mib`. The hardware cell names the GPU models recorded on the fit receipts and does not name a machine. GPU hours sum walls on which the receipt recorded a GPU process, and each DOPE fit file is counted once. Receipts with no fit timer are omitted from that table. CPU hours are omitted as well: no receipt stores a separate CPU timer. The logged replay medians 4.56 and 14.4 are not the elapsed time of the scored runs (not measured). Forest-Flow medians use the 12 native-grid operation walls. Its bytes are the 6 selected artifacts. 953 sums `fit_seconds` and 8,801 sums receipt operations. Neither sum is that median. Whole-device energy is not job energy. Gaussian copula, Chow-Liu, and independent-marginal attempts store one unsplit `wall_seconds`. Medians are 1.71, 0.168, and 0.162. Maxima are 39.9, 1.29, and 0.429. BeyondArena medians remain 5.00 and 14.0. Reading the main paper threshold counts from top to bottom, CatBoost is 97, 97, 97; linear is 93, 92, 90; and the MLP is 75, 75, 73.

H FIT CORESETS

The fit view is not the full PMLB table. Frozen catalog metadata draws each official training and test file by quota stratification on the target distribution and high-variance feature quantile bins, aiming at 1000 training rows and 1000 test rows, with at most 2000 rows in total and at least 101 rows. A source smaller than that quota keeps every row and still reserves a test share of at least 0.2. Of the 100 prepared lineages, 60 were already inside the quota. The other lineages use a hybrid coresets that keeps the observed feature and target range; the bin counts are 1 lineage at 11 bins, and 39 lineages at 50 bins. The generator then groups identical official-training rows and assigns those groups with seed 1729 so that about 80% of the rows become the fit view and the rest become the validation holdout. Duplicate rows are not split across that cut. On these lineages the fit view has 72 to 800 rows. The bound keeps every lineage on one worker view and leaves the official test sealed. It is not a second draw at scoring time.

I ARTIFACT HASHES

The rows below are SHA-256 of each ledger and round lock the paper reads, plus the commit and the file hash of the harness that measured the grouped-validation fidelity ledger. Regenerating the macros does not rewrite a ledger hash.

Ledger or lock	SHA-256
Code commit of the measured harness Harness SHA-256	17f3294d139e19f0ec831605f8750368eebc30bb a952062c0f83805f6442a440a5ae15293a843349 d423eaa106a9c56a20d14c0f
research/benchmark/results/paper-original-metadatas-v1/index.json	6d73d5f7b37849526a98a451b41a49e045ef21e9 11058d920940c5ea989f7fd9
docs/whitepaper/generated/original-field-map.json	5b164d434414d3389f639d3645f40b11a999b32c 56daa2cfe36d25b2205082b6
docs/whitepaper/scripts/paper.receipts.py	a2d6d45e3ff06a38ed61ab029fadfd14529978bc fc89618aa9ceb3fbe57238ee
research/benchmark/expanded.validation.metrics.py	a952062c0f83805f6442a440a5ae15293a843349 d423eaa106a9c56a20d14c0f
research/benchmark/methods.lock.json	899c355cb55769710ed369fea6b082589b318588 60fb74929bf4190de28aca33
research/benchmark/results/expanded-validation-diagnostics.json	cd20ac0d07a1dcab7c5e46db195adb463172b517 4387bb71a054b2e5f340beeb
research/benchmark/results/s3-coreset-procedure.json	73251e43cf7838687427569931bf6127f42e9323 74af33137c57fd8fe8544704
research/benchmark/results/s3-lineage-recording.json	bbd0852c49f595315104261efa4eaa6f50db257e 323b8c002aac6737a2dbe7d7
research/benchmark/results/density-matched-population-validation.json	5264b88a40ad5efb21d9b119789caeb13e71e911 a17f1d8b57f1d8e8ac31732a
research/benchmark/results/sdv-matched-population-validation.json	4dc367eb78159a0386882d23ae1f99b0aa2cc9f4 25dlaae499b6f3453222a44e
research/benchmark/results/s3-matched-forest-conformation-validation.json	97a25be902954cad16c5b5802d0e0ba468dea104 221efa2465434c94e6774a1f
research/benchmark/results/s3-data.lock.json	857b61324d0a5b7dbbb98e0c147fcc79c5bb8616 e7ca61afd6785469c17f038f
research/benchmark/results/arf-native-closure-watch-v1.receipt.json	7e8eeb1ba94ca747367e98d5a016d2ed3ed4c40c 5a98ca76d7a5fe27ce9c173c
research/benchmark/results/beyondarena-s3-inventory.json	504166764631dbdf5e02d116334af5cc587731fc f3efe130d62240e6dd512d64
docs/whitepaper/generated/fit-trace.json	d8c31b9e4eb59cb1a087cf5dcdd4a91a73eca00 f3704fbff647a7da6ce65522
docs/whitepaper/generated/loss-curves.json	0124915497c98e560ef99c1e5ecf7ad876f91744 39ff9c763cf1cf906b4aeb1
docs/whitepaper/generated/replay-cost.json	faf77e7ce1ed1d22127701bf68bbfe2f8d183f89 cbd3475dd80094888c36f086
docs/whitepaper/generated/provenance.json	dea64ea61eaacac0e287298bba3ba76aa7900819 aa00feb3a5d66cdcb4fb8a17
docs/whitepaper/generated/beyond-fit.json	a1447f908bc9c51876d345597039607c734b507b f85aa8e08db0639e80a1f036

J EXECUTABLE MFS-V3 CONFORMANCE

The release score uses the contract’s weighted geometric mean with a component floor: $100 \exp(\sum_i w_i \log \max(s_i, \epsilon) / \sum_i w_i)$, clipped to $[0, 100]$. The components lie in $[0, 1]$. A failed gate yields a null scalar. The matched null is the median linear-MSE candidate among three deterministic independent-marginal tables. Linear losses and candidate selection are fixed before any representation encoder forward. This is a median selector; a learnability band is outside this implementation. The near-copy floor is the declared quantile of nearest-other squared distances among distinct real fit rows in the shared map, with linear interpolation. Undefined floors fail the gate. The exact implementation and contract fields are:

Table 6: **Test-split provenance.** File-name counts only. No retention denominator and no Holm family.

Item	Record
Catalog SHA-256	ab9fda8d2dea46067b70a42812e9d3c1d9dc6ba025780100df34377e81aa1120
Eligible / prepared / overlap exclusions	104 / 100 / 4
Unknown-rights catalog entries held out	3755
Split	official_test_grouped_training_80_20, seed 1729
Overlap lineage ids	590210721973399a, 9fe58a656fdb458a, d06a584f07d0acd5, d0ba629ed07b5d63
Final evaluation authorized	false
S3 worker directories / train.csv / validation.csv	100 / 100 / 100
test.csv under S3 workers	0
BeyondArena workers / worker test.csv	5 / 0
Evaluator test.csv files, contents not read	5

Scalar	$100 \exp(\sum_i w_i \log \max(s_i, \epsilon) / \sum_i w_i)$, clipped to $[0, 100]$
Weights	compactness = 0.05, coverage_realism = 0.05, distribution_fidelity = 0.15, driver_fidelity = 0.15, representation_closeness = 0.15, structure_fidelity = 0.1, utility_transfer = 0.35
Epsilon	1×10^{-6}
Bytes	$b \in \mathbb{Z}$, $0 < b \leq 10240$
Prerequisites	Real-vs-real control; required sizes; artifact sampling; locked metric implementations.
Matched null	Median linear MSE of three independent-marginal candidates; candidate-index tie break.
Near-copy floor	$q = 0.01$ of nearest-other distinct real-row squared distances; linear interpolation.

The scorer tests in `research/benchmark/tests/test_mfs_v3.py` verify exact 100 for a perfect profile, the byte boundary, and each prerequisite. The panel tests verify median null selection and the quantile floor.

K PUBLIC DATA AND SCORED-ARTIFACT AVAILABILITY

A credential-free upstream-data path is `python3-mresearch.benchmark.fetch_pmlb--list` and `python3-mresearch.benchmark.fetch_pmlbDATASET_IDOUTPUT_DIRECTORY`. It authenticates PMLB data against an upstream Git LFS SHA pin at a fixed public revision. The public manifest includes every prepared lineage’s PMLB identifier and the historical train/validation/test split hashes from the committed data lock. Downloads stay upstream source data: the historical JopeDime staging used a separate hybrid coreset and target-stratified selection, whose exact selected row indices were omitted from the captured receipts. The public fetch therefore provides source datasets and historical split identities, while exact historical partition reconstruction requires the missing selection manifest.

The originally scored model files and synthetic rows have no published bundle. The captured historical metadata has artifact digests and charges but omits a complete portable inventory binding every scored file to a retrievable public object. Publishing current scratch files under those identities would require an additional custody audit; logged replay fits have different artifact identities. The committed validation ledgers, original metadata, and compressed numeric logs, authenticated by the input pins and rebuilt by `just paper`, reproduce the historical reductions and figures. The control, holdout-privacy, and TabSyn cell-level reductions remain in progress, wave 2. This paper prints no number for them. Original model sampling and historical partition reconstruction stay packaging limitations.

L DETAIL MOVED FROM THE MAIN TEXT

The sections below keep the long tables, the component definitions, and the delta-method expansion. A caveat here is the same caveat. The main text keeps the equations, the primary comparisons, and the release gate.

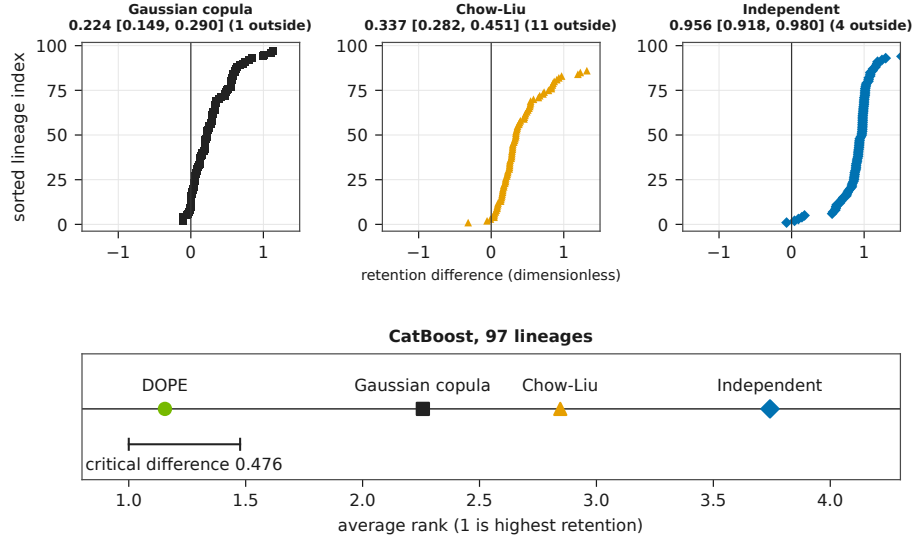


Figure 2: **CatBoost paired differences, sorted, and the Nemenyi scale.** The denominator is the sorted paired lineages. Rank 1 is the higher retention. The segment length is the critical difference. This display stays outside the nine-test Holm family.

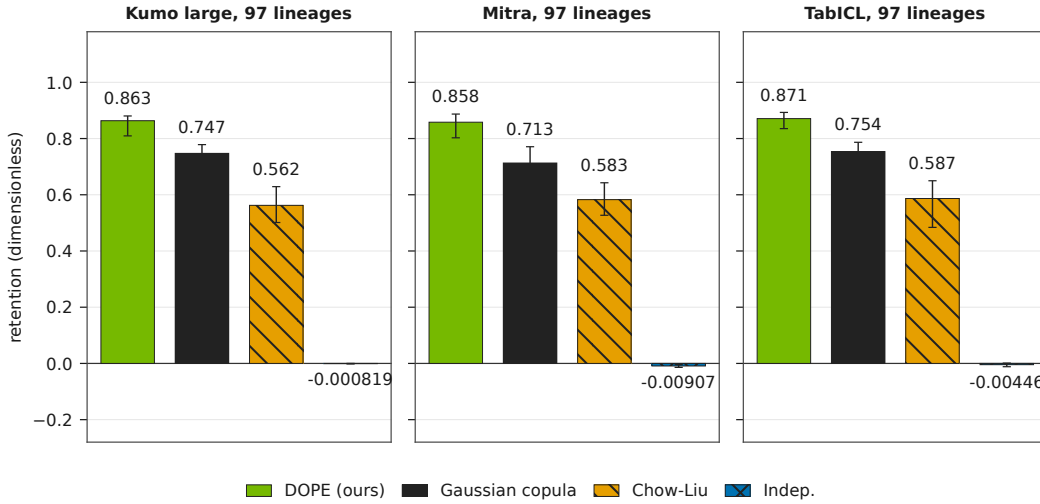


Figure 3: **Kumo large, Mitra, and TabICL retention at $4n$.** Higher is better. Each bar is the lineage median and its lineage-bootstrap interval. The denominator is the paired informative lineages in that panel. Hatches distinguish the methods. Holm family size is three per auditor and is reported in the main paper. This panel stays outside MFS.

L.1 RETAINED VALIDATION EVIDENCE

Retained validation. 8 and 9 add the merged retained-sample evidence. Each lineage averages all three sample seeds before a lineage-bootstrap median and percentile interval are computed. This differs from the median-of-seed rule in the earlier blocks; those blocks and their numbers are preserved. A lineage enters a metric only when all three values exist. An unavailable outcome stays absent. 8 covers the same prepared population for ARF, copula and Chow-Liu defaults and native-selected configurations. 9 includes only the complete scaled author-default TabSyn lineages and the classical methods on those identical inputs. It cannot establish a full-population TabSyn result or a native-tuned TabSyn comparison. These evaluations leave the configuration unselected.

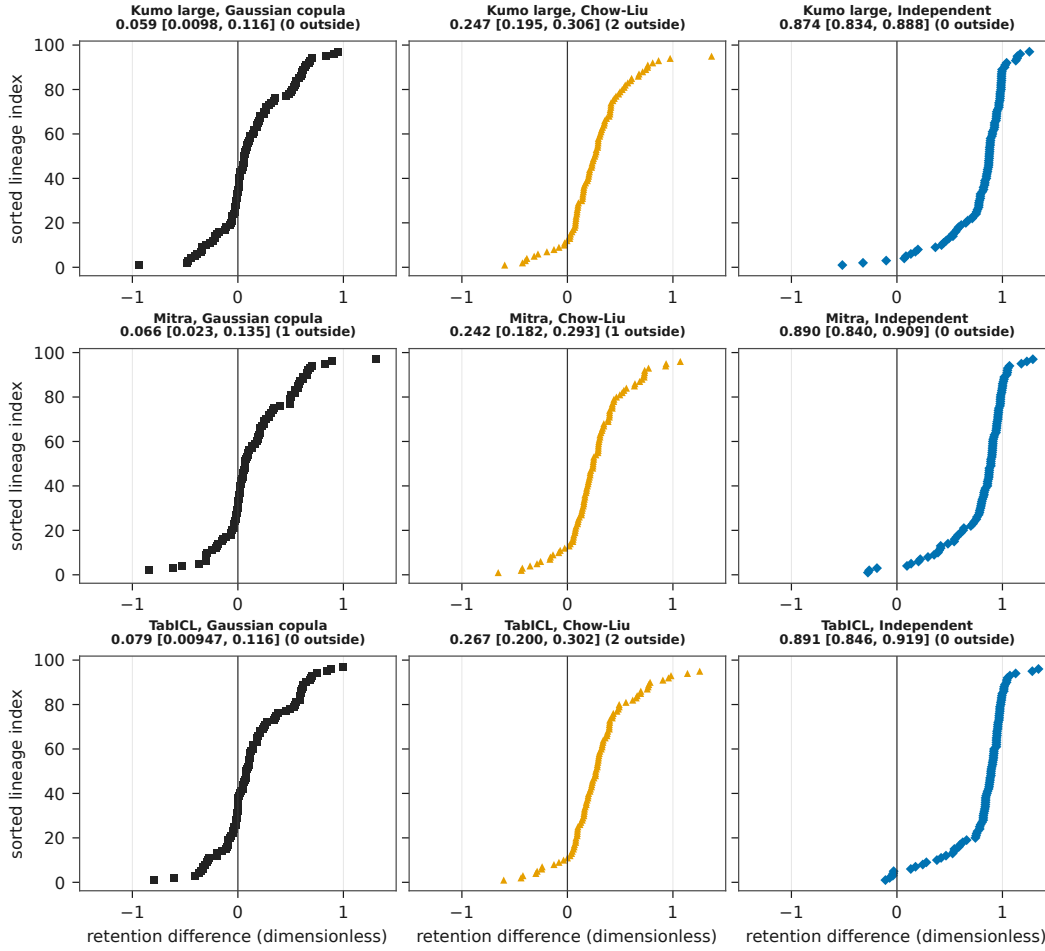


Figure 4: **Paired retention differences for Kumo large, Mitra, and TabICL, sorted by lineage.** Filled marks are lineages. The denominator is the paired informative lineages in that panel. Holm family size is three per auditor, in the main paper.

The matched TabSyn cohort remains in progress, wave 2, and this paper prints no number for it. That cohort stays distinct from this retained subset.

10 gives exploratory TabSyn-minus-comparator retention differences on shared lineages. The Holm family comprises four metrics (CatBoost retention, mean marginal KS/TV, CatBoost C2ST AUC and distance-MIA AUC) times six classical configurations. Conservative binomial/order-statistic intervals for the median difference use Bonferroni correction over the entire family, conditional on independent lineages. This small cohort requires unbounded simultaneous intervals. Finite tail-adjusted bootstrap ranges are retained only as exploratory descriptions; familywise coverage stays outside those descriptions. Marginal intervals in 5 stay short of simultaneous intervals. The machine-readable reduction includes every diagnostic contrast and all applicability exclusions. A raw AUC difference stays outside a directional utility or privacy win. Selection of the completed subset, one fit seed and its small lineage count limit this exploratory analysis.

11 separately extends the published DOPE-ARF retention comparison with lineage intervals. It preserves that receipt's median-of-three rule. Its six-test family contains two ARF configurations and three auditors. The shared-lineage intersections differ by auditor. These validation descriptions leave the release and superiority gates null.

The supplement prints the deterministic first-batch diagnostics per physical job. Incomplete seed groups and mixed research profiles prevent population confidence intervals or method rankings for that batch. All reductions trace to committed receipt paths, source digests and JSON pointers in

Table 7: **Grouped-validation fidelity and empirical privacy.** The fidelity denominator is the complete three-seed group count and stays distinct from the 97 retention lineages. Holm families are 12 density tests and 8 neural tests, separate from retention. Outside DP and HIPAA.

Method	α -prec. <i>Median [interval] (lineages).</i>	β -recall	DCR share (unbalanced)	DOMIAS
DOPE	0.905 [0.889, 0.918] (98)	0.437 [0.398, 0.473] (98)	0.811 [0.806, 0.814] (98)	0.508 [0.498, 0.520] (94)
Gaussian	0.812 [0.786, 0.832] (100)	0.430 [0.378, 0.453] (100)	0.806 [0.802, 0.811] (100)	0.513 [0.494, 0.527] (94)
Chow-Liu	0.858 [0.825, 0.904] (100)	0.359 [0.312, 0.400] (100)	0.810 [0.806, 0.816] (100)	0.512 [0.495, 0.522] (94)
Indep.	0.885 [0.868, 0.897] (100)	0.284 [0.197, 0.347] (100)	0.801 [0.797, 0.806] (100)	0.507 [0.496, 0.521] (94)
DOPE	0.902 [0.882, 0.940] (22)	0.441 [0.381, 0.489] (22)	0.811 [0.802, 0.819] (22)	0.508 [0.480, 0.538] (22)
CTGAN	0.719 [0.520, 0.856] (22)	0.170 [0.142, 0.267] (22)	0.820 [0.800, 0.842] (22)	0.502 [0.463, 0.534] (22)
TVAE	0.850 [0.731, 0.896] (22)	0.356 [0.283, 0.401] (22)	0.825 [0.813, 0.843] (22)	0.527 [0.487, 0.563] (22)

Contrast	α -prec.	β -recall	DCR share (unbalanced)	DOMIAS
<i>Line 2: Holm p; rank-biserial r; n. DOPE minus comparator.</i>				
	0.083 [0.045, 0.113]	0.024 [0.011, 0.039]	-0.0025 [-0.00625, 0.005]	0.00157 [-0.0069, 0.0074]
DOPE-Gaussian	1.86×10^{-10} ; 0.781; 98	0.394; 0.215; 98	1.000; 0.000206; 98	1.000; -0.085; 94
	0.019 [0.00782, 0.056]	0.048 [0.029, 0.072]	-0.00125 [-0.00562, 0.00125]	0.000581 [-0.00802, 0.011]
DOPE-Chow-Liu	2.00×10^{-5} ; 0.551; 98	1.32×10^{-10} ; 0.789; 98	0.806; -0.169; 98	1.000; 0.00572; 94
	0.018 [0.0072, 0.028]	0.104 [0.068, 0.138]	0.00625 [0.0025, 0.011]	0.00244 [-0.00595, 0.015]
DOPE-Indep.	5.47×10^{-5} ; 0.523; 98	9.71×10^{-15} ; 0.937; 98	0.005; 0.398; 98	1.000; 0.080; 94
	0.168 [0.054, 0.368]	0.219 [0.131, 0.256]	-0.00812 [-0.029, 0.015]	-0.00028 [-0.014, 0.031]
DOPE-CTGAN	0.004; 0.794; 22	3.81×10^{-6} ; 1.000; 22	0.532; -0.277; 22	0.610; 0.130; 22
	0.071 [0.00713, 0.141]	0.062 [0.00546, 0.135]	-0.025 [-0.030, -0.00641]	-0.021 [-0.036, -0.00972]
DOPE-TVAE	0.040; 0.644; 22	0.069; 0.573; 22	0.059; -0.613; 22	0.108; -0.510; 22

Table 8: **Retained classical validation at 4n.** Median [95% lineage-bootstrap interval] (informative lineages). The parenthetical count is the denominator. Native rows use the objective recorded for that configuration. No Holm family in this table.

Method	Config.	CB retention	KS/TV	C2ST CB AUC	Distance MIA AUC
ARF	Default	0.820 [0.787, 0.872] (99)	0.103 [0.097, 0.114] (100)	0.604 [0.582, 0.626] (100)	0.553 [0.538, 0.564] (100)
ARF	Native	0.806 [0.736, 0.855] (99)	0.105 [0.098, 0.114] (100)	0.610 [0.591, 0.627] (100)	0.542 [0.536, 0.559] (100)
Chow-Liu	Default	0.643 [0.552, 0.672] (99)	0.128 [0.106, 0.154] (100)	0.721 [0.652, 0.773] (100)	0.513 [0.503, 0.525] (100)
Chow-Liu	Native	0.566 [0.460, 0.631] (99)	0.126 [0.105, 0.157] (100)	0.702 [0.641, 0.761] (99)	0.513 [0.504, 0.523] (100)
Copula	Default	0.594 [0.427, 0.762] (99)	0.126 [0.094, 0.138] (100)	0.615 [0.591, 0.671] (100)	0.514 [0.503, 0.524] (100)
Copula	Native	0.641 [0.431, 0.803] (99)	0.134 [0.125, 0.158] (100)	0.730 [0.696, 0.779] (99)	0.507 [0.494, 0.521] (100)

docs/whitepaper/generated/retained-evidence.json. The original receipt references are retained in research/benchmark/results/retained-classical-validation/panel.json and the first-batch publication recorded in that reduction.

The coverage table distinguishes Forest-Flow confirmation from the Forest-Diffusion pilot modes. No full-population diffusion result is inferred from the Flow subset. Additional native-scored fits and published common-metric cells are counted separately. Pending publication withholds any claim that a method failed. Receipt paths and digests are recorded in docs/whitepaper/generated/baseline-coverage.json. The ARF follow-on now supplies complete five-fit common measurements on the same validation lineages. Its between-fit standard deviation is descriptive and stays outside a confidence interval. TabSyn’s additional retained n-sized evaluations include incomplete sample groups; those groups have no aggregate figure value. The supplementary descriptive figure inputs in docs/whitepaper/generated/published-baseline-kpis.csv retain a receipt path and JSON pointer for every value. Cohorts and fit counts differ; these inputs withhold a paired ranking. The ARF validation completion withholds final official-test certification.

L.2 BEYONDARENA

BeyondArena (2ecfe882ccfb814fc27c4de10a64ceefd5d7655c) stays outside the 100 PMLB lineages and stays unpooled from them.

5 of the 12 families in this fit check finished inside the cap. 7 stayed unprepared (4 outside the projected range, 3 overlapping). The range stayed at its recorded bound. Those 12 families are the

Table 9: **Matched retained TabSyn subset at 4n**. Same inputs and real-data controls for every method; median [95% interval] (lineages). The parenthetical count is the denominator. Separate from the full classical population. No Holm family in this table.

Method	Config.	CB retention	KS/TV	C2ST CB AUC	Distance MIA AUC
ARF	Default	0.839 [0.668, 0.942] (8)	0.129 [0.084, 0.165] (8)	0.623 [0.547, 0.969] (8)	0.555 [0.506, 0.613] (8)
ARF	Native	0.752 [0.534, 0.946] (8)	0.132 [0.084, 0.163] (8)	0.606 [0.583, 0.812] (8)	0.527 [0.482, 0.558] (8)
Chow-Liu	Default	0.746 [0.666, 0.859] (8)	0.186 [0.068, 0.255] (8)	0.756 [0.525, 0.999] (8)	0.522 [0.465, 0.542] (8)
Chow-Liu	Native	0.655 [0.561, 0.778] (8)	0.188 [0.087, 0.248] (8)	0.707 [0.525, 0.999] (8)	0.499 [0.470, 0.533] (8)
Copula	Default	0.643 [0.270, 0.935] (8)	0.146 [0.067, 0.242] (8)	0.687 [0.554, 0.987] (8)	0.510 [0.466, 0.541] (8)
Copula	Native	0.680 [0.260, 0.858] (8)	0.157 [0.097, 0.242] (8)	0.698 [0.642, 0.987] (8)	0.493 [0.463, 0.525] (8)
TabSyn	Scaled default	0.911 [0.685, 0.971] (8)	0.136 [0.120, 0.157] (8)	0.670 [0.597, 0.869] (8)	0.571 [0.547, 0.619] (8)

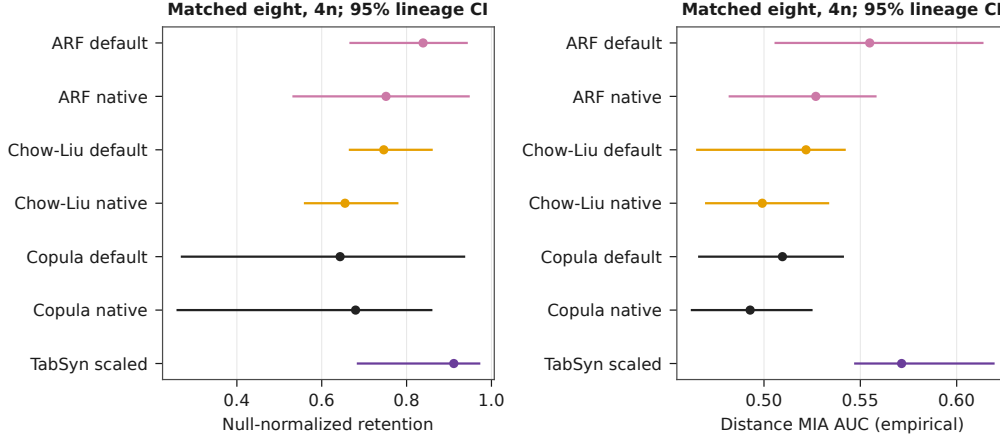


Figure 5: **Matched retained subset: median and marginal lineage-bootstrap intervals**. The denominator is the matched retained lineages in each panel. No Holm family. Privacy AUC is an empirical diagnostic; its interval stays outside a privacy guarantee.

fit check. The inventory is a separate list. The supplement names all 142 inventory families. 13 has no retention column.

M ABLATIONS

Step budget. `features12_steps8192` is the same residual with four times as many AdamW steps. It is a budget check and stays distinct from the displayed model. Where both budgets have a complete three-seed group, the CatBoost median paired difference is 0.034 (68 lineages). The supplement gives the linear and MLP differences and lists all 100 lineages. A blank cell is an unavailable fit or an uninformative auditor. The cell stays blank, and a zero stays unwritten. The scored fits leave the per-step series unstored. The replay section above plots the logged replay. The replay stays distinct from the scored artifacts. The lower loss at 8,192 steps leaves the displayed profile in place.

The primary informativeness threshold is 0.01 times the absolute null loss. 14 lists the CatBoost median by threshold and the informative counts. That Chow-Liu column uses its own informative count. The density-block Chow-Liu CatBoost median in the main paper is 0.586 on the paired informative lineages, so the two columns are different cohorts. The middle row is the 0.01 rule behind the retention tables. Thresholds 0, 0.001, and 0.05 keep that median and that count. At 0.1 the count is 96 and the median is 0.942. Fit-seed variance across seeds other than 11 is not measured for the displayed DOPE generator.

Distribution fidelity, structure fidelity, coverage realism, and compactness keep their v2 definitions. The driver term stored beside this panel is the same equal mixture of rank agreement and top-feature-set overlap, applied to absolute correlations with the target. It stays distinct from the permutation-importance auditor in the v2 definition, and this paper withholds it as a component

Table 10: **Exploratory matched-subset CatBoost differences.** The denominator is the shared lineages in the cell. Holm family size is four metrics times six configurations. Conservative simultaneous median intervals; the corrected intervals are unbounded. A non-significant test leaves equivalence unclaimed.

TabSyn minus	n	Median CB difference	Simultaneous median CI	Holm p
ARF/default	8	0.010	Unbounded	1.000
ARF/native	8	0.159	Unbounded	0.469
Chow-Liu/default	8	0.157	Unbounded	1.000
Chow-Liu/native	8	0.149	Unbounded	0.188
Copula/default	8	0.242	Unbounded	1.000
Copula/native	8	0.245	Unbounded	1.000

Table 11: **DOPE-ARF legacy matched receipt at $4n$: exploratory paired differences, conservative simultaneous median intervals and Holm p .** The denominator is the shared-lineage intersection for that auditor. Holm family size is six. Median-of-seed aggregation. A non-significant test leaves equivalence unclaimed.

ARF config.	Auditor	n	DOPE minus ARF	Simultaneous median CI	Holm p
Default	catboost	97	0.077	[0.019, 0.154]	1.31×10^{-5}
Native	catboost	97	0.095	[0.028, 0.175]	5.47×10^{-7}
Default	linear	92	0.027	[0.0056, 0.090]	8.53×10^{-5}
Native	linear	92	0.037	[0.00582, 0.113]	1.62×10^{-5}
Default	mlp	75	0.115	[0.043, 0.236]	4.56×10^{-8}
Native	mlp	75	0.143	[0.040, 0.253]	8.01×10^{-8}

value. v2 utility transfer remains the one-sided lower bound on retention for the CatBoost, linear, and MLP auditors in the tables below. v3 utility transfer is that same bound on the Kumo large retention. The model is conditioned on the synthetic rows and scored on real holdout rows. Mitra and TabICL must be measured on the same mapped tables for the gate. Their retentions stay out of that bound. A CatBoost retention cannot fill this component.

Real fit rows, real holdout rows, and synthetic rows share one map. The map is fit on the real fit rows only. Numeric columns go to the open unit interval by an empirical mid-rank quantile. Categorical levels are pre-encoded to fixed codes in that same interval. Column names become HMAC digests. The map withholds the minimum and the maximum. A second fit on the synthetic rows is a protocol failure. The tabular auditors and the embedding distance both see this mapped pair.

Representation closeness asks whether Kumo large places the mapped synthetic table near the mapped real table. The distance is half the Euclidean length between the two unit-length vectors, and it lies in $[0, 1]$. Closeness is one minus that distance divided by the distance to a matched null, clamped to $[0, 1]$. The synthetic table must also sit closer than an independent-marginal table of the same size, under the same map. The matched null is selected from three independent-marginal tables drawn on the shared map. The seed of each table is the first eight bytes, little-endian, of the SHA digest of the dataset, the method, the configuration, the sample seed, and the candidate index. The candidate indices are 0, 1, and 2. The loss is ordinary linear mean squared error on the mapped holdout. Candidates are ordered by loss, then by candidate index; the middle candidate is the matched null. Those losses are fixed before any encoder runs. Mitra and TabICL must show the same sign as Kumo large on the gap against the independent-marginal table. A sign flip, a zero match distance, or a missing vector makes the component null. The component is required. It stays separate from the tabular transfer.

The headline encoder is Kumo large. Its loss is the structured-data-model regressor’s mean of the inverse-transformed quantile forecasts. A single in-context prediction is a different scale and stays outside this loss. Mitra and TabICL repeat that transfer on the same mapped tables. Both are required for the tabular gate. Neither is averaged into representation closeness. Mitra is the zero-shot regressor. TabICL restores the context scale before the squared error. Kumo small and the project encoder a0 stay unscored. Kumo is available under the OpenMDW license and the Apache license. Mitra is Apache-licensed. TabICL is BSD-licensed. The chase score refuses Foundation and TabPFN, whose licenses stay outside the clean set for this release. It also refuses the target encoder, the historical context assembly, Hyperion, TabDPT, and the target-aware blocks.

Table 12: **Baseline coverage from committed publications.** The denominator is the committed publication named in the row. Pending rows have no estimated result or completion date. No Holm family.

Method	Published measured coverage	Remaining campaign
TabSyn	49 scaled-default lineages, 143 measured cells at fit11: eight complete $n/4n$ lineages (48 cells), plus 95 n cells on 41 lineages (19 complete/22 partial groups). Native tuning/full population pending.	Pending
TabDDPM	12 retained lineages; 126 physical/132 logical cells; 21 models, fit seed 11. Default/native cohorts differ; native search incomplete.	Pending
Forest-Flow	13 default lineages at five fit seeds; 65 models/390 cells. Separate legacy six-lineage default/native panel; full population pending.	Pending
Forest-Diffusion	Pilot only; separate 100-lineage matched diffusion panel pending.	Pending
ARF	100 default/native lineages at five fit seeds, $n/4n$, three sample seeds: 6,000 logical cells / 5,892 physical receipts (108 aliases). Between-fit SD available; official-test certification pending.	Pending
GaussianCopula	100 default/native lineages at fit seed 11; 1200 logical cells. Five-fit common coverage pending.	Pending
Chow-Liu	100 default/native lineages at fit seed 11; 1200 logical cells. Five-fit common coverage pending.	Pending

Table 13: **BeyondArena families that prepared.** The denominator is the 12-family fit check and stays distinct from the 142-family inventory. No retention auditor and no Holm family.

Family	Fit rows	2,048 B	8,192 B	Val 2,048	Val 8,192
airfoil	802	804	818	0.010	0.00448
garments	939	2,317	2,372	0.028	0.025
musk	3,738	9,273	9,350	0.146	0.090
telemonitoring	257	1,852	1,890	0.033	0.019
video	35,918	5,054	5,071	0.00247	0.00195

TabDPT’s weights are Apache-licensed and still refused, because that model was trained on real tables that overlap this panel. Hyperion is the project’s and still refused, because one training arm saw labels. The 4,168-dimensional vector remains the router embedding. It stays outside the historical formula. The shipped generator remains `features12.steps2048`.

The object a recipient keeps is that generator together with a hash manifest: a salt identifier, the normalizer name, the encoder pin, one digest per column, a dtype code, and the byte length. The manifest carries no row, header, category string, extremum, or encoder weight. The salt and the hash-to-label map stay in a local file the recipient holds. A public sidecar that echoes a source string or an extremum fails the cleartext gate. An exact copy of a real row fails its own gate. The near-copy floor is the 0.01 quantile, with linear interpolation, of squared Euclidean distances from each distinct real fit row to the nearest other distinct real fit row on the shared map. A synthetic row whose squared distance falls below that floor fails the gate. An undefined floor fails the gate. Crossing 10,240 bytes fails the byte gate. In each case the scalar stays null.

The tabular error and the embedding distance answer different questions. Tabular transfer asks whether a clean tabular model conditioned on the synthetic rows recovers the improvement that the same model obtained from real fit rows. Representation closeness asks whether the mapped synthetic table sits near the mapped real table. A release needs both measurements. A copied row, a cleartext string, or a byte-cap failure still removes the score.

N RELATED WORK

Generators, by family. Parametric copulas (Patki et al., 2016) and Chow–Liu trees (Chow & Liu, 1968) are the classical density models. Conditional GANs and variational autoencoders, including CTGAN, TVAE, TableGAN, CTAB-GAN, and CTAB-GAN+, are a second family (Xu et al., 2019; Park et al., 2018; Zhao et al., 2021; 2024). Diffusion and score-based models are a third family. TabDDPM and TabSyn sit in that family with STASy, CoDi, TabDiff, CDTD, Forest-Diffusion, and GATD (Kotelnikov et al., 2023; Zhang et al., 2024; Kim et al., 2023; Lee et al., 2023; Shi et al., 2025; Mueller et al., 2025; Jolicoeur-Martineau et al., 2024; Zagardo, 2026). Autoregressive and language-model generators, and prior-data fitted generators, are a fourth family (Borisov et al., 2023; Solatorio & Dupriez, 2023; Ma et al., 2023; Hollmann et al., 2025). Graphical neural generators are separate from those four (Liu et al., 2023). ARF estimates density

Table 14: **Informativeness threshold and auditor counts.** The denominator for each median is the informative count printed beside it. No Holm family.

Threshold	DOPE	Gaussian	Chow-Liu	Indep.
0	0.940 (97)	0.656 (99)	0.582 (99)	-0.012 (99)
0.001	0.940 (97)	0.656 (99)	0.582 (99)	-0.012 (99)
0.01	0.940 (97)	0.656 (99)	0.582 (99)	-0.012 (99)
0.05	0.940 (97)	0.656 (99)	0.582 (99)	-0.012 (99)
0.1	0.942 (96)	0.668 (98)	0.584 (98)	-0.012 (98)

Threshold	CatBoost	Linear	MLP
0.001	97	93	75
0.01	97	92	75
0.05	97	90	73

with a forest (Watson et al., 2023). CART synthesis is the synthpop line (Nowok et al., 2016). A survey collects these design choices (Borisov et al., 2024). SMOTE is an oversampling method and stays unrun (Chawla et al., 2002).

What this paper measured. The measured generators are the Gaussian copula, a Chow-Liu baseline written for this comparison, independent marginals written for this comparison, CTGAN, TVAE, ARF, a scaled author-default TabSyn subset (the retained-validation section of this supplement), and a 6-lineage Forest-Flow confirmation. TabSyn and TabDDPM are both in the diffusion family above. TabSyn’s measured cells are the retained subset, with a full population still unmeasured. The TabDDPM population publication is not measured. The other named generators are cited and stay unmeasured. the coverage table in this supplement records that coverage. This panel scores regression retention. Classification oversampling stays outside that score.

Utility ratios, fidelity, and privacy. Retention in the retention ratio is a null-normalized TSTR/TRTR ratio. Esteban, Hyland, and Rätsch named train-on-synthetic, test-on-real and train-on-real, test-on-real for synthetic series (Esteban et al., 2017). The numerator is the synthetic-trained auditor’s improvement over a null. The denominator is the real-trained auditor’s improvement over that null. The ratio is the measurement column. The claim in this paper is the byte-budgeted encoding and the release gate in the main paper, which stays null on every method in these tables. Utility reviews separate marginal fidelity from downstream prediction (Snoke et al., 2018; Raab et al., 2018). Alpha-precision, beta-recall, SDMetrics, and synthcity are the usual fidelity reports (Alaa et al., 2022; Zhang et al., 2020; Qian et al., 2023). Differentially private Bayesian networks and marginal synthesizers optimize a privacy budget (Zhang et al., 2014; McKenna et al., 2021; 2022; Cai et al., 2021). Membership inference and shadow-model attacks are a separate detection literature (Shokri et al., 2017; Hayes et al., 2018; van Breugel et al., 2023). Formal differential privacy is a different claim from an empirical resemblance score (Dwork & Roth, 2014; Abadi et al., 2016). The fidelity results add alpha-precision, beta-recall, a distance-to-closest-record share, and a DOMIAS AUC. ARF and the retained TabSyn subset have an additional fidelity, detection and empirical-privacy panel in the retained-validation section of this supplement. The displayed-profile ARF publication median is 0.840. The pre-specified privacy-attack panel remains not measured. Nothing in those fields is a formal privacy guarantee or a HIPAA de-identification claim.

Auditors and rank tests. Tree models remain strong tabular predictors, which is why CatBoost is one of the three auditors (Grinsztajn et al., 2022; Prokhorenkova et al., 2018; Shwartz-Ziv & Armon, 2022). TabZilla and TabArena show how brittle a single split can be (McElfresh et al., 2023; Erickson et al., 2026). The regression lineages come from PMLB (Olson et al., 2017; Romano et al., 2022). Comparing several methods on the same lineages needs a rank test and a critical difference, or a Bayesian signed-rank alternative (Demšar, 2006; Benavoli et al., 2017). This paper reports the frequentist procedure: a Friedman test and a Nemenyi critical difference on the common set, plus paired Wilcoxon tests. The Bayesian signed-rank procedure stays uncomputed.

O DISCUSSION

CTGAN and TVAE fall short of the CatBoost retention, and their successful size- $4n$ artifacts miss the cap. None of these comparisons emits MFS-v2, MFS-v3, PTF-v1, or a release at the 10,240-byte tier. The Kumo, Mitra, and TabICL table in this supplement is a retention table and leaves those scores unemitted. A small Wilcoxon p -value leaves that null in place. The 8,192-step budget moves the CatBoost median by 0.034 and stays distinct from the displayed model. BeyondArena fits 5 further families inside the cap and leaves retention unshown. The unmeasured publications and the null release scores are unchanged from the scope paragraph of the main text. An equal tuning budget across generators stays unmeasured.

O.1 WHAT THE SIX RELEASE-SCORE COMPONENTS MEASURE

Utility transfer is a one-sided lower bound on retention, clamped into $[0, 1]$. It is the bound inside the historical formula, and it is distinct from the median of the retention ratio in the figures of the main text. Driver fidelity is an equal mixture of feature-importance rank agreement and top-feature-set overlap between the real table and the synthetic table. Distribution fidelity is half the minimum marginal fidelity, plus a quarter of the sliced-Wasserstein fidelity and a quarter of the maximum mean discrepancy fidelity. Structure fidelity is the minimum dependence fidelity.

P PROCEDURES MOVED FROM THE MAIN TEXT

P.1 DATA LOCK, RETENTION ALGEBRA, AND RANK PROCEDURE

The data lock admits 104 MIT-licensed entries and prepares 100. 4 lineages have no split object because the same source-row digest appears in two partitions (`check_nonoverlap` raises on that overlap). Another 3,755 catalog entries lack a redistribution grant and stay out of the prepared set. The prepared cut is a grouped 80/20 of the official training rows, seed 1729. Worker directories hold `train.csv` and `validation.csv`. `test.csv` stays out of those directories. The fit view has 72 to 800 rows. Sources above the quota use a hybrid coreset that keeps the observed feature and target range. Held-out validation rows are the grouped holdout of the official training file. Test-file contents stay unread.

The ARF watch log’s last count is 799 closed and 799 ok out of 800 planned, then a sealing record.

Rearrangement of the retention ratio, with r the retention, is algebra on the three losses named in the main text:

$$L_{\text{synthetic}} = L_{\text{real}} + (1 - r)(L_{\text{null}} - L_{\text{real}}).$$

Invented losses 1, 0.01, and retention 0.94 give a synthetic-trained loss of 0.0694, which is 6.94 times the real-trained mean squared error. The CatBoost median in this paper is not this example.

On CatBoost the Holm sign-test values are 2.15×10^{-15} , 6.72×10^{-24} , and 4.95×10^{-27} . A Friedman test and a Nemenyi critical difference use the four density methods on their common complete-informative set, with rank 1 assigned to the higher retention.

The procedures below are the sampling draw, the lineage-name inventory, the fidelity readout, the compute timers, and the longer limitation record. A caveat here is the same caveat.

P.2 JOINT DRAW AND COMPILE SEARCH

The residual equation in the main text is the conditional mean. The joint draw is separate. Sampling for this profile uses the symbolic kernel stored in the artifact. The separate neural-joint sampler stays unused. The dependence stored with an artifact is one choice from the compile search: independence, a Chow–Liu tree, a triangular autoregression, a sparse Gaussian copula, a factor model, a truncated vine, or a two-component mixture. The displayed artifacts span several dependence families.

Training fits the residual. Sampling draws a row from the stored artifact. Training screens the inputs, ahead of sampling, by absolute rank correlation with the target and keeps at most 12. The artifact stores a marginal and a transform for each feature, the selected dependence, the residual, a

noise model, and a per-feature missingness rate. A draw samples a latent vector from the stored dependence, maps each coordinate through the standard normal distribution function, inverts the marginal, and inverts the feature transform. The residual then supplies $\hat{y}$. When the training target falls outside an all-zero or all-one column, the stored noise is heteroscedastic: a linear fit of the log absolute residual on up to four screened features, with the scale clamped between the lower and upper deciles of those absolute residuals. The sampled target adds Gaussian noise of that scale and then clamps the sum to the unit interval of the training transform. The returned target stays on that unit interval. Each feature is then replaced, independently, by a missing mark with its stored probability. For the displayed DOPE artifacts, the charged byte count is the encoded model file plus `projection.json`. The publisher adds those two lengths. A baseline charge is the length stored for that method. This paper keeps that stored length. Before that encoding, the compiler quantizes the marginals, the dependence, the residual, and the noise model.

The profile was fixed before the shared retention was inspected. Compile-time search records the proxy $0.45u + 0.25d + 0.30j$ and then keeps the candidate with the smallest normalized gate shortfall, then the smallest artifact, then the lowest candidate id. Every candidate record withholds a release certificate. The compile record stays outside a release certificate. No global DOPE family has been selected. The production default, used for a profile other than this research profile, clamps the hidden width from 4 to 12 and trains for 96 steps. That default stays outside this result. Other shapes stay outside this measurement: micro-TVAEs, TinyMAT, joint TVAE, a masked transformer, and a TabSyn candidate. A direct-rank TabDDPM candidate is a research comparator with no claim on the 10,240-byte tier. The scored runs leave the per-step loss unstored. A later logged replay stores one. The supplement plots that replay. The replay files stay distinct from the scored artifacts, and a lower hidden-basis loss at 8,192 steps leaves the displayed profile in place. The full width-and-step grid is not measured.

Of these prepared lineages, 20 names begin with feynman, which the PMLB catalog describes as synthetic physics, 14 begin with strogatz, which that catalog describes as simulated dynamics, 25 contain the Friedman-style fri marker, 6 contain bng and are Bayesian-network generators, and 35 are the remainder. This paper does not re-verify which of the remainder are observational measurements. Distinct lineage identifiers are not a claim of independent source families. Related names, including the `cpu_small` lineages, stay inside the one-row-per-lineage bootstrap. The resampling unit is the lineage, so related variants move together only by chance. A source-family interval is not published, because a verified parent map is not in the ledger.

Fidelity and empirical privacy. On the density cells, the displayed profile has median classifier two-sample AUC 0.517 (interval 0.504 to 0.532, $n = 98$), mean marginal Kolmogorov–Smirnov distance 0.114 (0.103 to 0.135, $n = 98$), and pairwise correlation fidelity 0.953 (0.945 to 0.961, $n = 98$). Each of those counts is the number of lineages with a complete three-seed fidelity group. It stays distinct from the 97 informative CatBoost retention lineages and from 100. A two-sample AUC near one half means a linear-time classifier has trouble telling the synthetic rows from the held-out validation rows. The supplement adds alpha-precision, beta-recall, the unbalanced DCR share, and a DOMIAS AUC. The lower block is the neural lineages only. Holm families are 12 and 8, separate from retention. The DCR column is the fraction of synthetic queries whose nearest reference is in the fit view, with ties scored one half. A balanced variant was computed in the harness and stays outside this column. Under an independent draw, the unbalanced share sits near the fit-view fraction of the pooled references. The grouped split keeps about four fifths of those rows in the fit view, so the baseline is near four fifths, not one half. Higher values of that share, and higher DOMIAS AUC, sit nearer the fit view and stay short of a quality win. DOMIAS here is the isolated kernel density-ratio equation from the pinned author expression, evaluated on this numeric projection. It stays distinct from the original DOMIAS training pipeline and from a membership-inference certificate. The DOMIAS AUC stays near one half. The density DCR contrast against independent marginals clears Holm in that worse direction. The displayed-profile ARF publication median is 0.840. The pre-specified privacy-attack panel remains not measured. This stays outside differential privacy and outside HIPAA de-identification.

The supplement’s compute table lists fit time, peak memory, the GPU models, GPU hours, and charged bytes for receipts that store a fit timer. Sample-timer medians (maxima) are 186.7 (197.0) for CTGAN, 186.1 (196.6) for TVAE, and 342.4 (511.7) for Forest-Flow. The DOPE receipt stores no sample timer. No receipt stores a separate CPU timer. Gaussian copula, Chow–Liu, independent marginals, and ARF have no fit timer, so they are omitted; their byte medians are in the

supplement. ARF receipts leave the GPU model unrecorded. TabSyn cost is not measured (the retained validation subset reports quality, not cost). TabDDPM cost is not measured (this paper scores no TabDDPM population quality). The supplement states the replay walls, the Forest-Flow sums, and the unsplit density attempt timers.

Internal. The primary displayed DOPE and legacy comparison fits use the coreset in the experimental setup in the main text and fit seed 11, and a training-derived validation cut of the official training rows. The scored retention panel is regression only. Other tasks stay outside the result of these tables. The supplement records the draw. The official test stayed out of training, selection, and the numbers in this paper. The registered endpoint partition is `untouched_real_test`. This paper scores the training-derived validation cut and leaves that partition sealed. The deviation is disclosed here. The result remains a validation estimate. The informativeness rule drops a negligible real-versus-null gap. Thresholds 0, 0.001, and 0.05 keep that median and that count. At 0.1 the count is 96 and the median is 0.942. Incomplete groups are omitted. A zero for an omitted group stays unwritten. Chow–Liu and independent marginals are implementations written for this comparison. CTGAN and TVAE use the lineages that finished. Runs that stopped at the recorded time limit are omitted. They stay outside the failure count of a finished fit.

The supplementary budget protocol retains the frozen 600-second Forest-Flow fit cap and all timed-out attempt receipts. Timeout cells have missing quality metrics and confer no DOPE win; coverage denominators include their recorded dispositions. The October 8 disposition addendum was written after 75 observed timeouts and stays short of retroactive preregistration. A longer-budget sensitivity track would require a separate prospective freeze and comparable method budgets, with both tracks’ costs retained. The supplement’s `BASELINE_PROTOCOL.md` links the frozen completion checkpoint and every timeout disposition.

The validation cut also supplies native tuning signals. These are post-selection validation estimates. Independent official-test estimates stay sealed. Lineage intervals hold the split and fitted artifact fixed; they omit fitting uncertainty. Related simulation lineages can violate the independence assumption behind both bootstrap and order-statistic intervals. The retained-subset results therefore remain exploratory.

External. The primary displayed DOPE fits leave behavior on the full tables used by the diffusion papers, across five fit seeds, and on the official test unestablished. The additional ARF and Forest-Flow fit seeds leave DOPE fitting uncertainty unestablished. Generalization past seed 1729 is not measured for the displayed DOPE generator. Forest-Flow’s byte comparison uses 6 lineages and the hash-checked fit files. BeyondArena is a separate fit check. It stays outside the retention panel and stays unpooled from the density block.

Construct. Retention is the reported utility. Both release scores in the release-score section of the main text stay null. The linear auditor leaves DOPE unseparated from the Gaussian copula, and equivalence stays unclaimed. Shipping this artifact stays a different experiment from shipping the compact predictor itself. The predictor-only control and the real-row resample remain in progress, wave 2. This paper prints no number for them. Equivalence stays unclaimed. Normalized-space RMSE is the loss the auditors stored. Original-unit error after inverting the training transform stays unstored, and this paper leaves it unreported. The supplement’s fidelity table stays outside a membership-inference certification. Its unbalanced DCR share and DOMIAS AUC are empirical. Empirical resemblance stays outside differential privacy and outside HIPAA de-identification.

Statistical. The rank tests are descriptive. The pre-registered production superiority test stays null. Density, neural, Forest-Flow, and BeyondArena stay unpooled. A small retention p -value leaves the null scalar in place. A non-significant test leaves equivalence unclaimed. The supplement gives the delta-method variance of one lineage’s retention and the reason a small denominator is dropped. The intervals in the tables remain the lineage bootstrap of the median. Those intervals omit the delta-method term.

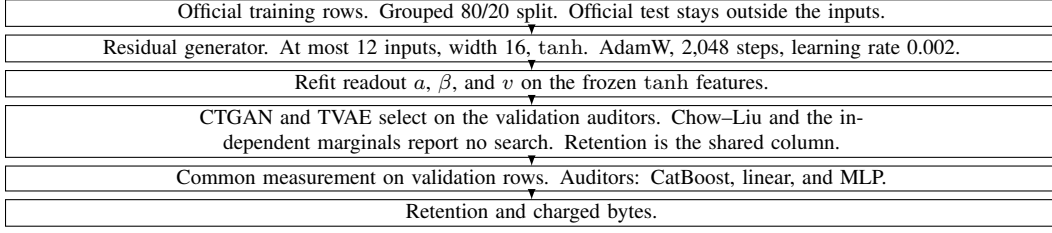


Figure 6: **Displayed pipeline.** The official test stays outside the inputs. No lineage denominator and no Holm family.

P.3 TRAINING FREEZE, ROUTER VECTOR, AND EMISSION REQUIREMENTS

Training screens the inputs, maps them through the width-16 tanh layer, and runs AdamW on the regression mean-squared error, with gradient clipping at 5. The hidden weights and biases are then frozen. The intercept, the linear skip, and the hidden readout are refit on those frozen features. Coefficients whose absolute value is below 10^{-5} are dropped from the skip.

The 4,168-dimensional vector appends 24 candidate states from one router bundle onto the 856-wide sketch in the main text. A second bundle moves the same table. An index stores the 856-wide sketch. The longer vector is a route from one bundle. A neighbor found with the sketch is still fit with the displayed profile. The artifact is still charged in bytes, and it still faces the release score in the main text. The sketch stays outside the fitness components. A router action that names a retention or a failure probability is a prediction from the bundle. It stays outside the historical formula.

PTF-v1 hard gates include joint fidelity, membership and attribute attacks, copy checks, and feature-importance agreement on a sealed holdout. PTF-v1 is null here, as are MFS-v2 and MFS-v3. Compile-time search records the proxy $0.45u + 0.25d + 0.30j$ and then keeps the candidate with the smallest gate shortfall, then the smallest artifact. That record withholds a release certificate. The proxy stays outside both release-score formulas.

Emitting MFS-v2 requires the six components, a non-empty hard-gate map that passed, both the size- n and size- $4n$ artifacts, real-versus-real controls, artifact-only sampling, and locked metric implementations. Emitting MFS-v3 requires the bounded formula in the main text: the four prerequisites named there, the clean tabular transfer, the embedding distance under the shared map, and a pass on every hard gate. The Kumo, Mitra, and TabICL table reports the measured retentions. A retention stays distinct from the scalar. A retention win leaves those inputs empty. Copying a retention number into the historical formula would publish a value the contract refuses to emit. The generated rows apply the bounded formula to invented components, including a zero compactness term. Those rows are arithmetic on the bounded formula. They show the shape of the product. CTGAN and TVAE select on the validation auditors. Chow–Liu and the independent marginals report no search. The shared columns in this paper stay retention and charged bytes.

Q TABLES AND FIGURES MOVED FROM THE MAIN TEXT

The displays below are the selection rule, the compute ledger, and the block tables named from the main text. A caveat here is the same caveat. The main text keeps the headline table, the bytes figure, and the teaser.

RMSE is the square root of the stored mean squared error. Training-mean skill is one minus that loss divided by the training-mean null loss. Conventional held-out R^2 uses the mean of the evaluation targets and stays unstored for these rows. Retention is the median of the per-lineage ratios and stays distinct from the ratio of these medians. CatBoost reads RMSE 0.053 and 0.081, with training-mean skill 0.881 and 0.792.

The two displayed DOPE artifacts over the cap are 4544_GeographicalOriginalofMusic at 22,233 bytes and 588_fri_c4_1000_100 at 19,019 bytes. 678_visualizing_environmental has a fit whose CatBoost group stays uninformative, so the CatBoost median uses 97 lineages while the byte count

Table 15: **Selection rule for each method.** The scored column is retention on the validation cut. CTGAN and TVAE use the same validation rows and the same linear and MLP auditors as that score. No Holm family. The budget column is the rule recorded for the method. A matched compute ledger stays outside that column.

Method	Criterion	Data used	Budget
DOPE displayed	Profile fixed before the shared retention was read. Compile search keeps the smallest gate shortfall, then the smallest artifact, then the lowest candidate id. Proxy $0.45u + 0.25d + 0.30j$.	Training-derived validation cut. Official test sealed.	At most 12 inputs, width 16, 2,048 steps, seed 11.
Gaussian copula	Held-out mean log density. Copulas 0.14.1 GaussianMultivariate.	Same validation cut. Distinct from the retention auditors.	One library configuration.
CTGAN	Mean of linear and MLP training-mean skill.	Validation rows and those two auditors, the same ones used at scoring.	Author implementation. Stopped runs omitted.
TVAE	Same skill as CTGAN.	Same rows and auditors as CTGAN.	Author implementation. Stopped runs omitted.
Chow–Liu	No search.	Fit view, eight bins, Laplace smoothing 1.0. Written for this comparison.	Fixed.
Independent marginals	No search.	Fit-view marginals. Written for this comparison.	Fixed.
ARF	Author-default and native-objective rows stay separate. Median 0.840.	Matched validation publication. Left on that receipt.	See the publication receipt.
Forest-Flow	Author regression objective on a frozen grid.	6 confirmation lineages. Separate block.	Fit cap 600 seconds.
TabSyn	No configuration selected.	Scaled author default on the complete lineages in the supplement.	Cost is not measured (the retained validation subset reports quality, not cost).
TabDDPM	No scored population selection.	Publication is not measured.	Cost is not measured (this paper scores no TabDDPM population quality).

Table 16: **Fit time, memory, GPU, and charged bytes where a fit timer was stored.** The denominator is the receipts that stored a timer. No Holm family.

Method	Fit s	RAM / VRAM	Hardware	GPU h	Bytes
DOPE	7.25(13.5)	920.7/ 1067	one NVIDIA GeForce RTX 4090 and two NVIDIA GeForce RTX 3090 GPUs	0.193	1,895.5(758–22,233)
CTGAN	213.5(268.1)	1557.8/ 1101	NVIDIA GeForce RTX 4090	2.52	741,834(528,075–1,428,695)
TVAE	214.2(253.2)	1252.9/ 1069	NVIDIA GeForce RTX 4090	2.45	317,141.5(157,578–741,659)
Forest-Flow	185.9(380.6)	6112.0/ 3181	NVIDIA GeForce RTX 4090	2.44	325,848,913.5(115,202,253–1,085,155,062)

uses 100. Independent marginals pass the cap more often than the Chow–Liu baseline and still sit near retention zero. Smallness alone stays outside the result. The displayed byte range runs from 758 to 22,233.

R REVIEW PANELS

The panels below come from the pinned review receipts. A method with two configurations keeps a separate row for each configuration. Every filled TOST cell in the paired-difference table is no. Equivalence stays unclaimed. A missing artifact stays out of the median. Empirical distance and membership scores stay outside differential privacy and outside HIPAA de-identification. Four generated files stay out of this section because an ARF block repeats there without an author-default versus native-selected column: the TOST-only file, the RMSE file, the stored logistic fidelity file, and the existing-seed file.

R.1 PAIRED DIFFERENCES AT BOTH SIZES

The Wilcoxon and TOST calculations treat paired lineages as independent. The printed interval is the family-cluster bootstrap of the median. Holm is split by size. Each size has nine density tests, six neural tests, six ARF tests, and three Forest-Flow tests. The Holm column in this table is printed at three decimals. The main-text headline table prints the same ARF values in scientific notation.

Size	Auditor	Comparator	Configuration	n	Median diff.	Hierarchical CI	Mean	TOST	W/T/L	Holm p
1n	catboost	GaussianCopula	native_selected	97	0.2178	[0.1028, 0.3340]	0.2483	no	90/0/7	0.000

1n	catboost	Chow-Liu	native_selected	97	0.3635	[0.2986, 0.5314]	7.7460	no	93/0/4	0.000
1n	catboost	independent_marginals	native_selected	97	0.9238	[0.8575, 1.0178]	1.1580	no	97/0/0	0.000
1n	linear	GaussianCopula	native_selected	92	0.0088	[-0.0118, 0.0329]	0.2382	no	53/0/39	0.071
1n	linear	Chow-Liu	native_selected	92	0.3309	[0.2275, 0.6482]	1.5301	no	86/0/6	0.000
1n	linear	independent_marginals	native_selected	92	0.9955	[0.9691, 1.0693]	1.1054	no	92/0/0	0.000
1n	mlp	GaussianCopula	native_selected	75	0.1453	[0.0479, 0.2445]	0.1971	no	64/0/11	0.000
1n	mlp	Chow-Liu	native_selected	75	0.5339	[0.3343, 0.8171]	1.1319	no	75/0/0	0.000
1n	mlp	independent_marginals	native_selected	75	1.0462	[0.9834, 1.2772]	1.3698	no	75/0/0	0.000
4n	catboost	GaussianCopula	native_selected	97	0.2237	[0.0723, 0.3530]	0.2613	no	86/0/11	0.000
4n	catboost	Chow-Liu	native_selected	97	0.3370	[0.2823, 0.5048]	8.0937	no	94/0/3	0.000
4n	catboost	independent_marginals	native_selected	97	0.9556	[0.8911, 0.9875]	1.2876	no	96/0/1	0.000
4n	linear	GaussianCopula	native_selected	92	0.0037	[-0.0144, 0.0138]	0.1832	no	49/0/43	0.373
4n	linear	Chow-Liu	native_selected	92	0.3399	[0.1947, 0.6595]	1.4415	no	86/0/6	0.000
4n	linear	independent_marginals	native_selected	92	0.9879	[0.9427, 1.0142]	1.0440	no	92/0/0	0.000
4n	mlp	GaussianCopula	native_selected	75	0.1850	[0.0811, 0.2624]	0.3295	no	65/0/10	0.000
4n	mlp	Chow-Liu	native_selected	75	0.5301	[0.3183, 0.8832]	0.9903	no	74/0/1	0.000
4n	mlp	independent_marginals	native_selected	75	1.1043	[1.0367, 1.3283]	1.3100	no	75/0/0	0.000
1n	catboost	CTGAN	native_selected	21	0.8816	[0.7166, 1.4415]	1.0872	no	21/0/0	0.000
1n	catboost	TVAE	native_selected	21	0.2381	[0.1133, 0.4819]	0.2867	no	19/0/2	0.000
1n	linear	CTGAN	native_selected	21	1.0516	[0.8722, 1.3740]	1.4244	no	21/0/0	0.000
1n	linear	TVAE	native_selected	21	0.2510	[0.1501, 0.4434]	1.3010	no	20/0/1	0.000
1n	mlp	CTGAN	native_selected	19	0.9336	[0.6818, 2.0568]	2.1780	no	19/0/0	0.000
1n	mlp	TVAE	native_selected	19	0.2725	[0.1706, 0.7092]	0.5985	no	18/0/1	0.000
4n	catboost	CTGAN	native_selected	21	0.8807	[0.7020, 1.3326]	1.0962	no	21/0/0	0.000
4n	catboost	TVAE	native_selected	21	0.2361	[0.0868, 0.4471]	0.2673	no	19/0/2	0.000
4n	linear	CTGAN	native_selected	21	1.0485	[0.8718, 1.4642]	1.3489	no	21/0/0	0.000
4n	linear	TVAE	native_selected	21	0.2629	[0.1298, 0.4183]	1.0124	no	20/0/1	0.000
4n	mlp	CTGAN	native_selected	19	1.0788	[0.8605, 2.0296]	1.8412	no	19/0/0	0.000
4n	mlp	TVAE	native_selected	19	0.2842	[0.1227, 0.8120]	0.7771	no	17/0/2	0.000
1n	catboost	ARF	author_default	97	0.1086	[0.0026, 0.2394]	0.1144	no	73/0/24	0.000
1n	catboost	ARF	native_selected	97	0.1010	[0.0236, 0.2404]	0.1354	no	76/0/21	0.000
1n	linear	ARF	author_default	92	0.0409	[0.0049, 0.1752]	0.1786	no	64/0/28	0.000
1n	linear	ARF	native_selected	92	0.0649	[0.0059, 0.2012]	0.2123	no	68/0/24	0.000
1n	mlp	ARF	author_default	75	0.1371	[0.0311, 0.2860]	0.2632	no	61/0/14	0.000
1n	mlp	ARF	native_selected	75	0.1423	[0.0464, 0.2643]	0.3097	no	64/0/11	0.000
4n	catboost	ARF	author_default	97	0.0767	[0.0103, 0.1957]	0.2059	no	69/0/28	0.000
4n	catboost	ARF	native_selected	97	0.0954	[0.0166, 0.2137]	0.1737	no	70/0/27	0.000
4n	linear	ARF	author_default	92	0.0274	[-0.0020, 0.1195]	0.0924	no	61/0/31	0.000
4n	linear	ARF	native_selected	92	0.0369	[0.0008, 0.1639]	0.1249	no	65/0/27	0.000
4n	mlp	ARF	author_default	75	0.1153	[0.0340, 0.3709]	0.2894	no	57/0/18	0.000
4n	mlp	ARF	native_selected	75	0.1429	[0.0283, 0.3673]	0.3089	no	58/0/17	0.000
1n	catboost	ForestDiffusion/Forest-Flow	native_selected	6	-0.0198	[-0.4355, 0.0060]	-0.0917	no	1/0/5	0.469
1n	linear	ForestDiffusion/Forest-Flow	native_selected	5	0.0275	[-0.1396, 0.0723]	-0.0040	no	3/0/2	1.000
1n	mlp	ForestDiffusion/Forest-Flow	native_selected	4	0.0196	[-0.0405, 0.2906]	0.0723	no	2/0/2	1.000
4n	catboost	ForestDiffusion/Forest-Flow	native_selected	6	-0.0269	[-0.4550, 0.0080]	-0.1089	no	1/0/5	0.375
4n	linear	ForestDiffusion/Forest-Flow	native_selected	5	0.0115	[-0.2599, 0.3016]	0.0118	no	3/0/2	0.812
4n	mlp	ForestDiffusion/Forest-Flow	native_selected	4	-0.0317	[-0.2275, -0.0102]	-0.0753	no	0/0/4	0.375

R.2 CONTROLS, HOLDOUT PRIVACY, AND THE TABSYN COHORT

The predictor-only control, the real-row resample, the holdout privacy panel, and the matched TabSyn cohort remain in progress, wave 2. This paper prints no number for them. The retained TabSyn subset above is a different cohort. The stored logistic detector stays out of this section.

R.3 SOURCE-FAMILY SENSITIVITY

Per-source-family medians and an excluding-simulated cut. The interval is the hierarchical CI. These rows are a sensitivity and stay unpooled from the headline.

Family	Size	Auditor	Contrast	n	Median	Hierarchical CI
feynman	1n	catboost	DOPE	20	0.9703	[0.9454, 0.9906]
feynman	1n	catboost	GaussianCopula	20	0.2002	[0.1097, 0.2549]
feynman	1n	catboost	Chow-Liu	20	0.3253	[0.2200, 0.4252]
feynman	1n	catboost	independent_marginals	20	1.0201	[0.9776, 1.0254]
feynman	1n	linear	DOPE	20	0.9938	[0.9806, 0.9975]
feynman	1n	mlp	DOPE	19	0.9713	[0.9609, 0.9885]
feynman	4n	catboost	DOPE	20	0.9894	[0.9713, 0.9944]
feynman	4n	catboost	GaussianCopula	20	0.1894	[0.0993, 0.2840]
feynman	4n	catboost	Chow-Liu	20	0.2829	[0.2346, 0.3576]
feynman	4n	catboost	independent_marginals	20	0.9813	[0.9777, 0.9947]
feynman	4n	linear	DOPE	20	0.9941	[0.9854, 0.9982]
feynman	4n	mlp	DOPE	19	0.9965	[0.9860, 1.0067]
strogatz	1n	catboost	DOPE	14	0.7819	[0.4900, 0.9793]
strogatz	1n	catboost	GaussianCopula	14	0.1925	[-0.0086, 0.3450]

strogatz	1n	catboost	Chow-Liu	14	0.8451	[0.2765, 2.3318]
strogatz	1n	catboost	independent_marginals	14	0.8779	[0.6389, 1.0237]
strogatz	1n	linear	DOPE	13	0.9687	[0.8345, 0.9960]
strogatz	1n	mlp	DOPE	9	0.9967	[0.8086, 1.0923]
strogatz	4n	catboost	DOPE	14	0.8854	[0.5626, 0.9845]
strogatz	4n	catboost	GaussianCopula	14	0.1303	[-0.0086, 0.3896]
strogatz	4n	catboost	Chow-Liu	14	0.8739	[0.2276, 2.9286]
strogatz	4n	catboost	independent_marginals	14	0.9465	[0.6883, 1.0851]
strogatz	4n	linear	DOPE	13	0.9718	[0.7502, 0.9969]
strogatz	4n	mlp	DOPE	9	1.0815	[0.9862, 1.4876]
fri	1n	catboost	DOPE	24	0.8292	[0.7550, 0.8791]
fri	1n	catboost	GaussianCopula	24	0.4997	[0.3340, 0.5624]
fri	1n	catboost	Chow-Liu	24	0.3401	[0.2105, 0.4416]
fri	1n	catboost	independent_marginals	24	0.8660	[0.7905, 0.8976]
fri	1n	linear	DOPE	23	0.9675	[0.8787, 1.0009]
fri	1n	mlp	DOPE	15	0.9889	[0.8894, 1.0310]
fri	4n	catboost	DOPE	24	0.9054	[0.8485, 0.9276]
fri	4n	catboost	GaussianCopula	24	0.5583	[0.3530, 0.5913]
fri	4n	catboost	Chow-Liu	24	0.2964	[0.1981, 0.4561]
fri	4n	catboost	independent_marginals	24	0.9152	[0.8394, 0.9537]
fri	4n	linear	DOPE	23	0.9960	[0.9668, 1.0359]
fri	4n	mlp	DOPE	15	1.5792	[1.0250, 1.6797]
bng	1n	catboost	DOPE	6	0.9118	[0.0978, 1.1020]
bng	1n	catboost	GaussianCopula	6	0.1041	[0.0472, 0.1614]
bng	1n	catboost	Chow-Liu	6	0.1781	[0.0312, 0.6353]
bng	1n	catboost	independent_marginals	6	1.0364	[0.5939, 1.3518]
bng	1n	linear	DOPE	5	0.9718	[0.9305, 1.0382]
bng	1n	mlp	DOPE	4	1.0475	[0.9751, 1.6088]
bng	4n	catboost	DOPE	6	0.9334	[0.2662, 1.0918]
bng	4n	catboost	GaussianCopula	6	0.0387	[-0.0122, 0.1742]
bng	4n	catboost	Chow-Liu	6	0.0761	[0.0430, 0.5724]
bng	4n	catboost	independent_marginals	6	0.9060	[0.4007, 1.2632]
bng	4n	linear	DOPE	5	0.9941	[0.9173, 1.0458]
bng	4n	mlp	DOPE	4	1.2962	[0.9871, 1.7049]
other	1n	catboost	DOPE	33	0.9020	[0.8417, 0.9725]
other	1n	catboost	GaussianCopula	33	0.1651	[0.0633, 0.2597]
other	1n	catboost	Chow-Liu	33	0.4339	[0.2997, 0.6052]
other	1n	catboost	independent_marginals	33	0.9217	[0.8303, 0.9973]
other	1n	linear	DOPE	31	0.9812	[0.9713, 0.9952]
other	1n	mlp	DOPE	28	0.9322	[0.8545, 0.9864]
other	4n	catboost	DOPE	33	0.9373	[0.8908, 0.9878]
other	4n	catboost	GaussianCopula	33	0.1912	[0.0762, 0.2495]
other	4n	catboost	Chow-Liu	33	0.4514	[0.3214, 0.6687]
other	4n	catboost	independent_marginals	33	0.9293	[0.8699, 1.0026]
other	4n	linear	DOPE	31	0.9951	[0.9662, 1.0017]
other	4n	mlp	DOPE	28	1.0955	[1.0357, 1.1953]
excluding simulated	1n	catboost	GaussianCopula	33	0.1651	[0.0698, 0.2637]
excluding simulated	1n	catboost	Chow-Liu	33	0.4339	[0.2997, 0.6052]
excluding simulated	1n	catboost	independent_marginals	33	0.9217	[0.8575, 0.9973]
excluding simulated	4n	catboost	GaussianCopula	33	0.1912	[0.0762, 0.2495]
excluding simulated	4n	catboost	Chow-Liu	33	0.4514	[0.2957, 0.6687]
excluding simulated	4n	catboost	independent_marginals	33	0.9293	[0.8699, 1.0026]

R.4 INFORMATIVE-LINEAGE DENOMINATORS

Informative, noninformative, and undefined lineage counts. The sensitivity column is the capped cohort. A different denominator is a different cohort.

Method	Configuration	Size	Informative	Noninformative	Undefined	Capped	Sensitivity n	Capped median
DOPE	features12_steps2048	1n	97	1	0	1	98	0.8862
DOPE	features12_steps2048	4n	97	1	0	1	98	0.9359
GaussianCopula	native_selected	1n	99	1	0	1	100	0.5996
GaussianCopula	native_selected	4n	99	1	0	1	100	0.6658
Chow-Liu	native_selected	1n	99	1	0	12	100	0.4809
Chow-Liu	native_selected	4n	99	1	0	11	100	0.5782
independent_marginals	native_selected	1n	99	1	0	3	100	-0.0440
independent_marginals	native_selected	4n	99	1	0	1	100	-0.0120
CTGAN	native_selected	1n	21	1	0	2	22	-0.0889
CTGAN	native_selected	4n	21	1	0	2	22	-0.0801
TVAE	native_selected	1n	21	1	0	0	22	0.6094
TVAE	native_selected	4n	21	1	0	0	22	0.6582
ARF	author_default	1n	99	1	0	1	100	0.7505
ARF	author_default	4n	99	1	0	1	100	0.8314
ARF	native_selected	1n	99	1	0	1	100	0.7410
ARF	native_selected	4n	99	1	0	1	100	0.8237
ForestDiffusion/Forest-Flow	native_selected	1n	0	0	0	0	0	—

S DELTA-METHOD VARIANCE OF RETENTION

Let $D = L_{\text{null}} - L_{\text{real}}$ and write the retention ratio as $R = (L_{\text{null}} - L_{\text{synthetic}})/D$ when $D \neq 0$. The partial derivatives are

$$\begin{aligned}\frac{\partial R}{\partial L_{\text{null}}} &= \frac{L_{\text{synthetic}} - L_{\text{real}}}{D^2}, \\ \frac{\partial R}{\partial L_{\text{synthetic}}} &= -\frac{1}{D}, \\ \frac{\partial R}{\partial L_{\text{real}}} &= \frac{R}{D}.\end{aligned}$$

If the three losses have covariance Σ , the delta-method variance of this one lineage's retention is $\nabla R^\top \Sigma \nabla R$. The inverse powers of D in this gradient allow the variance to diverge as D approaches zero. The informativeness rule therefore drops a lineage whose real improvement is zero or smaller than 0.01 times the absolute null loss. The rule is a stability cut. A published standard error stays outside this rule. This appendix states the expansion. A numerical covariance stays unplugged, and the lineage-bootstrap intervals in the tables stay in place.

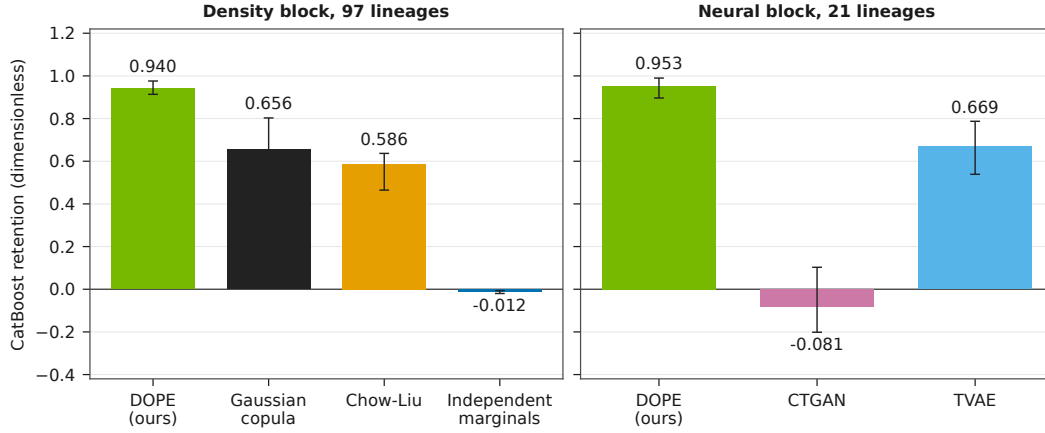


Figure 7: **CatBoost retention at $4n$.** Higher is better. DOPE names the encoding’s generator. Each bar is a method’s paired-lineage retention median and its lineage-bootstrap interval. Density uses 97 lineages. The neural panel uses 21 and stays unpooled. Forest-Flow, 6 lineages, is higher on its own block and is omitted. No Holm family is drawn. The tests are in the tables.

Table 17: **Density block at $4n$.** Each cell is a dimensionless retention. The denominator is the paired informative lineages in that row. Holm family size is nine. W/T/L counts lineages. A tie is a difference of zero. Equivalence stays unclaimed.

Auditor	Comparator	n	DOPE	Other	Difference	W/T/L	Holm p
CatBoost	Gaussian copula	97	0.940 [0.914, 0.976]	0.656 [0.431, 0.803]	0.224 [0.149, 0.290]	86/0/11	1.35×10^{-13}
CatBoost	Chow-Liu	97	0.940 [0.914, 0.976]	0.586 [0.465, 0.637]	0.337 [0.282, 0.451]	94/0/3	4.85×10^{-16}
CatBoost	Indep. marginals	97	0.940 [0.914, 0.976]	-0.012 [-0.020, -0.00649]	0.956 [0.918, 0.980]	96/0/1	1.17×10^{-16}
Linear	Gaussian copula	92	0.994 [0.981, 0.996]	0.989 [0.968, 0.994]	0.00366 [-0.00438, 0.012]	49/0/43	0.373
Linear	Chow-Liu	92	0.994 [0.981, 0.996]	0.661 [0.356, 0.738]	0.340 [0.244, 0.425]	86/0/6	7.12×10^{-15}
Linear	Indep. marginals	92	0.994 [0.981, 0.996]	0.00206 [-0.023, 0.00583]	0.988 [0.975, 1.001]	92/0/0	5.70×10^{-16}
MLP	Gaussian copula	75	1.061 [1.024, 1.125]	0.928 [0.865, 0.953]	0.185 [0.128, 0.228]	65/0/10	4.96×10^{-11}
MLP	Chow-Liu	75	1.061 [1.024, 1.125]	0.692 [0.582, 0.754]	0.530 [0.350, 0.651]	74/0/1	2.11×10^{-13}
MLP	Indep. marginals	75	1.061 [1.024, 1.125]	-0.028 [-0.047, -0.013]	1.104 [1.047, 1.179]	75/0/0	2.11×10^{-13}

Table 18: **Kumo large, Mitra, and TabICL on the density block at $4n$.** Each cell is a dimensionless retention and stays outside MFS. The denominator is the paired informative lineages in that row. Holm family size is three per auditor, separate from the nine-test family in the statistics section of the main paper. W/T/L counts lineages. A tie is a difference of zero. Equivalence stays unclaimed.

Auditor	Comparator	n	DOPE	Other	Difference	W/T/L	Holm p
Kumo large	Gaussian copula	97	0.863 [0.809, 0.880]	0.747 [0.668, 0.778]	0.059 [0.0098, 0.116]	62/0/35	5.70×10^{-4}
Kumo large	Chow-Liu	97	0.863 [0.809, 0.880]	0.562 [0.501, 0.629]	0.247 [0.195, 0.306]	85/0/12	1.34×10^{-11}
Kumo large	Indep. marginals	97	0.863 [0.809, 0.880]	-0.000819 [-0.00189, 7.58e-05]	0.874 [0.834, 0.888]	94/0/3	7.46×10^{-17}
Mitra	Gaussian copula	97	0.858 [0.803, 0.887]	0.713 [0.585, 0.771]	0.066 [0.023, 0.135]	65/0/32	1.34×10^{-4}
Mitra	Chow-Liu	97	0.858 [0.803, 0.887]	0.583 [0.527, 0.643]	0.242 [0.182, 0.293]	85/0/12	2.71×10^{-11}
Mitra	Indep. marginals	97	0.858 [0.803, 0.887]	-0.00907 [-0.014, -0.00377]	0.890 [0.840, 0.909]	94/0/3	6.01×10^{-17}
TabICL	Gaussian copula	97	0.871 [0.835, 0.893]	0.754 [0.672, 0.787]	0.079 [0.00947, 0.116]	62/0/35	1.61×10^{-4}
TabICL	Chow-Liu	97	0.871 [0.835, 0.893]	0.587 [0.484, 0.650]	0.267 [0.200, 0.302]	87/0/10	7.12×10^{-12}
TabICL	Indep. marginals	97	0.871 [0.835, 0.893]	-0.00446 [-0.012, 0.0012]	0.891 [0.846, 0.919]	92/0/5	5.83×10^{-17}

Holm family: 3 tests per auditor; DOPE versus Gaussian copula, Chow-Liu, and independent marginals.

Table 19: **Raw auditor medians on the informative lineages.** The denominator is the informative lineages for that auditor. No Holm family.

Auditor	n	RMSE real	RMSE synth.	training-mean skill, real	training-mean skill, synth.
CatBoost	97	0.053	0.081	0.881	0.792
Linear	92	0.105	0.112	0.585	0.551
MLP	75	0.098	0.083	0.666	0.785

Table 20: **Charged bytes at $4n$.** For displayed DOPE artifacts the charge is the encoded model file plus `projection.json`. The denominator is the artifacts with a stored charge. CatBoost retention uses 97 lineages. Forest-Flow stays unpooled. No Holm family.

Method	Median bytes	Within cap
DOPE reference	1,895.5	98/100
Gaussian copula	5,464.5	62/100
Chow-Liu	29,865	31/100
Independent marginals	2,549.5	94/100
CTGAN	741,834	0/22
TVAE	317,141.5	0/22
Forest-Flow	325,848,913.5	0/6
ARF	1,627,718.5	0/100

Table 21: **Neural block at $4n$, unpooled.** The denominator is the paired informative lineages in the cell. CatBoost uses 21. Holm family size is six. W/T/L counts lineages. A tie is a difference of zero. Equivalence stays unclaimed.

Auditor	Comparator	n	DOPE	Other	Difference	W/T/L	Holm p
CatBoost	CTGAN	21	0.953 [0.896, 0.990]	-0.081 [-0.202, 0.103]	0.881 [0.702, 1.150]	21/0/0	5.72×10^{-6}
CatBoost	TVAE	21	0.953 [0.896, 0.990]	0.669 [0.539, 0.787]	0.236 [0.087, 0.431]	19/0/2	1.57×10^{-4}
Linear	CTGAN	21	0.999 [0.996, 1.015]	-0.033 [-0.225, 0.080]	1.049 [0.916, 1.236]	21/0/0	5.72×10^{-6}
Linear	TVAE	21	0.999 [0.996, 1.015]	0.796 [0.605, 0.867]	0.263 [0.166, 0.410]	20/0/1	1.57×10^{-4}
MLP	CTGAN	19	1.026 [0.998, 1.263]	-0.149 [-0.471, 0.083]	1.079 [0.877, 2.030]	19/0/0	1.53×10^{-5}
MLP	TVAE	19	1.026 [0.998, 1.263]	0.753 [0.478, 0.852]	0.284 [0.153, 0.734]	17/0/2	1.57×10^{-4}

Table 22: **Forest-Flow confirmation at $4n$, CatBoost.** The denominator is 6 lineages. Holm family size is three. Unpooled. A non-significant test leaves equivalence unclaimed.

Lineage	DOPE bytes	DOPE	Forest-Flow bytes	Forest-Flow
1193_BNG_lowbwt	2,436	0.970	416,921,273	1.004
503_wind	3,846	0.996	637,056,689	1.016
582_fri_c1_500_25	5,670	0.909	1,085,155,062	0.883
596_fri_c2_250_5	1,698	0.807	234,776,554	0.968
feynman_I.12_1	1,032	0.991	123,784,681	1.001
strogatz_vdp1	984	0.581	115,202,253	1.036

Table 23: **Published ARF matched receipts at $4n$, CatBoost.** The denominator is the paired complete informative lineages in the generated row. No Holm family in this table.

Publication	Lineages	DOPE	Receipt	Difference
ARF, author_default	97	0.940	0.840	0.077
ARF, native_selected	97	0.940	0.826	0.095

REFERENCES

- Martin Abadi, Andy Chu, Ian Goodfellow, H. Brendan McMahan, Ilya Mironov, Kunal Talwar, and Li Zhang. Deep learning with differential privacy. In *Proceedings of the 2016 ACM SIGSAC Conference on Computer and Communications Security*, pp. 308–318, 2016. doi: 10.1145/2976749.2978318.
- Ahmed Alaa, Boris Van Breugel, Evgeny S. Saveliev, and Mihaela van der Schaar. How faithful is your synthetic data? sample-level metrics for evaluating and auditing generative models. In *Proceedings of the 39th International Conference on Machine Learning*, volume 162 of *Proceedings of Machine Learning Research*, pp. 290–306. PMLR, 2022. doi: 10.48550/ARXIV.2102.08921. URL <https://proceedings.mlr.press/v162/alaa22a.html>.
- Alessio Benavoli, Giorgio Corani, Janez Demšar, and Marco Zaffalon. Time for a change: a tutorial for comparing multiple classifiers through bayesian analysis. *Journal of Machine Learning Research*, 18(77):1–36, 2017. URL <https://jmlr.org/papers/v18/16-305.html>.
- Vadim Borisov, Kathrin Seßler, Tobias Leemann, Martin Pawelczyk, and Gjergji Kasneci. Language models are realistic tabular data generators. In *International Conference on Learning Representations*, 2023. doi: 10.48550/ARXIV.2210.06280. URL <https://openreview.net/forum?id=cEygmQNOeI>.
- Vadim Borisov, Tobias Leemann, Kathrin Seßler, Johannes Haug, Martin Pawelczyk, and Gjergji Kasneci. Deep neural networks and tabular data: A survey. *IEEE Transactions on Neural Networks and Learning Systems*, 35(6):7499–7519, 2024. doi: 10.1109/tnnls.2022.3229161.
- Kuntai Cai, Xiaoyu Lei, Jianxin Wei, and Xiaokui Xiao. Data synthesis via differentially private markov random fields. *Proceedings of the VLDB Endowment*, 14(11):2190–2202, 2021. doi: 10.14778/3476249.3476272.
- N. V. Chawla, K. W. Bowyer, L. O. Hall, and W. P. Kegelmeyer. SMOTE: Synthetic minority over-sampling technique. *Journal of Artificial Intelligence Research*, 16:321–357, 2002. doi: 10.1613/jair.953.
- C. K. Chow and C. N. Liu. Approximating discrete probability distributions with dependence trees. *IEEE Transactions on Information Theory*, 14(3):462–467, May 1968. doi: 10.1109/TIT.1968.1054142.
- Janez Demšar. Statistical comparisons of classifiers over multiple data sets. *Journal of Machine Learning Research*, 7(1):1–30, 2006. URL <https://jmlr.org/papers/v7/demsar06a.html>.
- Cynthia Dwork and Aaron Roth. The algorithmic foundations of differential privacy. *Foundations and Trends in Theoretical Computer Science*, 9(3–4):211–487, 2014. doi: 10.1561/04000000042.
- Nick Erickson, Lennart Purucker, Andrej Tschalzev, David Holzmüller, Prateek Mutalik Desai, David Salinas, and Frank Hutter. TabArena: A living benchmark for machine learning on tabular data. In *The Thirty-ninth Annual Conference on Neural Information Processing Systems Datasets and Benchmarks Track*, pp. 17285–17350, 2026. doi: 10.52202/085713-0519. URL <https://openreview.net/forum?id=jZqCqpCLdU>.
- Cristóbal Esteban, Stephanie L. Hyland, and Gunnar Rätsch. Real-valued (medical) time series generation with recurrent conditional GANs. *arXiv preprint arXiv:1706.02633*, 2017. doi: 10.48550/ARXIV.1706.02633.
- Leo Grinsztajn, Edouard Oyallon, and Gael Varoquaux. Why do tree-based models still outperform deep learning on typical tabular data? In *Advances in Neural Information Processing Systems*, volume 35, pp. 507–520, 2022. doi: 10.52202/068431-0037.
- Jamie Hayes, Luca Melis, George Danezis, and Emiliano De Cristofaro. LOGAN: Membership inference attacks against generative models. *Proceedings on Privacy Enhancing Technologies*, 2019(1):133–152, 2018. doi: 10.2478/popets-2019-0008.

- Noah Hollmann, Samuel Müller, Lennart Purucker, Arjun Krishnakumar, Max Körfer, Shi Bin Hoo, Robin Tibor Schirrmeister, and Frank Hutter. Accurate predictions on small data with a tabular foundation model. *Nature*, 637(8045):319–326, 2025. doi: 10.1038/s41586-024-08328-6.
- Alexia Jolicoeur-Martineau, Kilian Fatras, and Tal Kachman. Generating and imputing tabular data via diffusion and flow-based gradient-boosted trees. In *Proceedings of The 27th International Conference on Artificial Intelligence and Statistics*, volume 238 of *Proceedings of Machine Learning Research*, pp. 1288–1296. PMLR, 2024. doi: 10.48550/ARXIV.2309.09968. URL <https://proceedings.mlr.press/v238/jolicoeur-martineau24a.html>.
- Jayoung Kim, Chaejeong Lee, and Noseong Park. STaSy: Score-based tabular data synthesis. In *International Conference on Learning Representations*, 2023. doi: 10.48550/ARXIV.2210.04018. URL https://openreview.net/forum?id=lmNssCWt_v.
- Akim Kotelnikov, Dmitry Baranchuk, Ivan Rubachev, and Artem Babenko. TabDDPM: Modelling tabular data with diffusion models. In *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*, pp. 17564–17579, 2023. doi: 10.48550/ARXIV.2209.15421. URL <https://proceedings.mlr.press/v202/kotelnikov23a.html>.
- Chaejeong Lee, Jayoung Kim, and Noseong Park. CoDi: Co-evolving contrastive diffusion models for mixed-type tabular synthesis. In *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*, pp. 18940–18956. PMLR, 2023. doi: 10.48550/ARXIV.2304.12654. URL <https://proceedings.mlr.press/v202/lee23i.html>.
- Tennison Liu, Zhaozhi Qian, Jeroen Berrevoets, and Mihaela van der Schaar. GOGGLE: Generative modelling for tabular data by learning relational structure. In *International Conference on Learning Representations*, 2023. URL <https://openreview.net/forum?id=fPVRcJqspu>.
- Junwei Ma, Apoorv Dankar, George Stein, Guangwei Yu, and Anthony Caterini. TabPFGen – tabular data generation with TabPFN. In *NeurIPS 2023 Second Table Representation Learning Workshop*, 2023. URL <https://openreview.net/forum?id=4MkkNsAEoO>.
- Duncan McElfresh, Sujay Khandagale, Jonathan Valverde, Vishak Prasad C, Ganesh Ramakrishnan, Micah Goldblum, and Colin White. When do neural nets outperform boosted trees on tabular data? In *Advances in Neural Information Processing Systems*, volume 36, pp. 76336–76369, 2023. doi: 10.52202/075280-3337. URL https://proceedings.neurips.cc/paper_files/paper/2023/hash/f06d5ebd4ff40b40dd97e30cee632123-Abstract-Datasets_and_Benchmarks.html. Datasets and Benchmarks Track.
- Ryan McKenna, Gerome Miklau, and Daniel Sheldon. Winning the NIST contest: A scalable and general approach to differentially private synthetic data. *Journal of Privacy and Confidentiality*, 11(3), 2021. doi: 10.29012/jpc.778.
- Ryan McKenna, Brett Mullins, Daniel Sheldon, and Gerome Miklau. AIM: An adaptive and iterative mechanism for differentially private synthetic data. *Proceedings of the VLDB Endowment*, 15(11):2599–2612, 2022. doi: 10.14778/3551793.3551817.
- Markus Mueller, Kathrin Gruber, and Dennis Fok. Continuous diffusion for mixed-type tabular data. In *International Conference on Learning Representations*, 2025. doi: 10.48550/ARXIV.2312.10431. URL <https://openreview.net/forum?id=QPtoBPn4lZ>.
- Beata Nowok, Gillian M. Raab, and Chris Dibben. synthpop: Bespoke creation of synthetic data in R. *Journal of Statistical Software*, 74(11):1–26, 2016. doi: 10.18637/jss.v074.i11.

- Randal S. Olson, William La Cava, Patryk Orzechowski, Ryan J. Urbanowicz, and Jason H. Moore. PMLB: a large benchmark suite for machine learning evaluation and comparison. *BioData Mining*, 10(1):36, 2017. doi: 10.1186/s13040-017-0154-4.
- Noseong Park, Mahmoud Mohammadi, Kshitij Gorde, Sushil Jajodia, Hongkyu Park, and Youngmin Kim. Data synthesis based on generative adversarial networks. *Proceedings of the VLDB Endowment*, 11(10):1071–1083, 2018. doi: 10.14778/3231751.3231757.
- Neha Patki, Roy Wedge, and Kalyan Veeramachaneni. The synthetic data vault. In *2016 IEEE International Conference on Data Science and Advanced Analytics (DSAA)*, pp. 399–410, 2016. doi: 10.1109/DSAA.2016.49.
- Liudmila Prokhorenkova, Gleb Gusev, Aleksandr Vorobev, Anna Veronika Dorogush, and Andrey Gulin. CatBoost: unbiased boosting with categorical features. In *Advances in Neural Information Processing Systems*, volume 31, 2018. doi: 10.48550/ARXIV.1706.09516. URL <https://papers.nips.cc/paper/2018/hash/14491b756b3a51daac41c24863285549-Abstract.html>.
- Zhaozhi Qian, Rob Davis, and Mihaela van der Schaar. Synthcity: A benchmark framework for diverse use cases of tabular synthetic data. In *Advances in Neural Information Processing Systems*, volume 36, pp. 3173–3188, 2023. doi: 10.52202/075280-0140. URL https://proceedings.neurips.cc/paper_files/paper/2023/hash/09723c9f291f6056fd1885081859c186-Abstract-Datasets_and_Benchmarks.html. Datasets and Benchmarks Track.
- Gillian M. Raab, Beata Nowok, and Chris Dibben. Practical data synthesis for large samples. *Journal of Privacy and Confidentiality*, 7(3):67–97, 2018. doi: 10.29012/jpc.v7i3.407.
- Joseph D. Romano, Trang T. Le, William La Cava, John T. Gregg, Daniel J. Goldberg, Praneel Chakraborty, Natasha L. Ray, Daniel Himmelstein, Weixuan Fu, and Jason H. Moore. PMLB v1.0: an open-source dataset collection for benchmarking machine learning methods. *Bioinformatics*, 38(3):878–880, 2022. doi: 10.1093/bioinformatics/btab727.
- Juntong Shi, Minkai Xu, Harper Hua, Hengrui Zhang, Stefano Ermon, and Jure Leskovec. TabDiff: a mixed-type diffusion model for tabular data generation. In *International Conference on Learning Representations*, 2025. doi: 10.48550/ARXIV.2410.20626. URL <https://openreview.net/forum?id=swvURjrt8z>.
- Reza Shokri, Marco Stronati, Congzheng Song, and Vitaly Shmatikov. Membership inference attacks against machine learning models. In *2017 IEEE Symposium on Security and Privacy (SP)*, pp. 3–18, 2017. doi: 10.1109/sp.2017.41.
- Ravid Shwartz-Ziv and Amitai Armon. Tabular data: Deep learning is not all you need. *Information Fusion*, 81:84–90, 2022. doi: 10.1016/j.inffus.2021.11.011.
- Joshua Snoke, Gillian M. Raab, Beata Nowok, Chris Dibben, and Aleksandra Slavkovic. General and specific utility measures for synthetic data. *Journal of the Royal Statistical Society Series A: Statistics in Society*, 181(3):663–688, 2018. doi: 10.1111/rssa.12358.
- Aivin V. Solatorio and Olivier Dupriez. REaLTabFormer: Generating realistic relational and tabular data using transformers, 2023.
- Boris van Breugel, Hao Sun, Zhaozhi Qian, and Mihaela van der Schaar. Membership inference attacks against synthetic data through overfitting detection. In *Proceedings of The 26th International Conference on Artificial Intelligence and Statistics*, volume 206 of *Proceedings of Machine Learning Research*, pp. 3493–3514. PMLR, 2023. doi: 10.48550/ARXIV.2302.12580. URL <https://proceedings.mlr.press/v206/breugel23a.html>.
- David S. Watson, Kristin Blesch, Jan Kapar, and Marvin N. Wright. Adversarial random forests for density estimation and generative modeling. In *Proceedings of the 26th International Conference on Artificial Intelligence and Statistics*, volume 206 of *Proceedings of Machine Learning Research*, pp. 5357–5375, 2023. doi: 10.48550/ARXIV.2205.09435. URL <https://proceedings.mlr.press/v206/watson23a.html>.

- Lei Xu, Maria Skoularidou, Alfredo Cuesta-Infante, and Kalyan Veeramachaneni. Modeling tabular data using conditional GAN. In *Advances in Neural Information Processing Systems*, volume 32, 2019. doi: 10.48550/ARXIV.1907.00503. URL <https://papers.nips.cc/paper/2019/hash/254ed7d2de3b23ab10936522dd547b78-Abstract.html>.
- David Turtora Zagardo. Geometry-aware tabular diffusion. In *Proceedings of the 43rd International Conference on Machine Learning*, volume 306 of *Proceedings of Machine Learning Research*, pp. 152812–152835. PMLR, 2026. doi: 10.48550/ARXIV.2606.02607. URL <https://proceedings.mlr.press/v306/zagardo26a.html>.
- Hengrui Zhang, Jiani Zhang, Balasubramaniam Srinivasan, Zhengyuan Shen, Xiao Qin, Christos Faloutsos, Huzefa Rangwala, and George Karypis. Mixed-type tabular data synthesis with score-based diffusion in latent space. In *International Conference on Learning Representations*, 2024. doi: 10.48550/ARXIV.2310.09656. URL <https://openreview.net/forum?id=4Ay23yeuz0>.
- Jun Zhang, Graham Cormode, Cecilia M. Procopiuc, Divesh Srivastava, and Xiaokui Xiao. PrivBayes: Private data release via bayesian networks. In *Proceedings of the 2014 ACM SIGMOD International Conference on Management of Data*, pp. 1423–1434, 2014. doi: 10.1145/2588555.2588573.
- Kevin Alex Zhang, Kalyan Veeramachaneni, and Carles Sala. SDMetrics: Synthetic data metrics. Open-source library, MIT Data To AI Lab, 2020. URL <https://docs.sdv.dev/sdmetrics/>.
- Zilong Zhao, Aditya Kunar, Robert Birke, and Lydia Y. Chen. CTAB-GAN: Effective table data synthesizing. In *Proceedings of The 13th Asian Conference on Machine Learning*, volume 157 of *Proceedings of Machine Learning Research*, pp. 97–112. PMLR, 2021. URL <https://proceedings.mlr.press/v157/zhao21a.html>.
- Zilong Zhao, Aditya Kunar, Robert Birke, Hiek Van der Scheer, and Lydia Y. Chen. CTAB-GAN+: Enhancing tabular data synthesis. *Frontiers in Big Data*, 6, 2024. doi: 10.3389/fdata.2023.1296508.